MIL-M-28787/33B(EC)
4 February 1980
SUPERSEDING
MIL-M-28787/33A(EC)
28 February 1977

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SHIFT REGISTER, DIGITAL, KEY CODE JDJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic shift register module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II

Name	Shift register
Family	Digital
Key code	JDJ
Size	1A
Weight	35 grams maximum
Failure rate	0.33 failures/10^6 hours maximum
* Environment	Class I as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10 percent
Maximum power dissipation	1.6 W
Speed	DC to 16 MHz
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
A, B, C, D and SER inputs	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
M and CL inputs	
Low level (V_{IL} = 0 V)	13.8 mA maximum
High level (V_{IH} = 2.4 V)	320 µA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output voltage	
Low level	0.4 V maximum, 0 V minimum
High level	2.4 V minimum, V_{CC} maximum
Maximum output drive capability	
Low level (V_{OL} = 0.4 V)	16 mA maximum
High level (V_{OH} = 2.4 V)	-800 µA minimum
* Propagation delay time (CL to Q)	
t_{PLH}	6 ns minimum, 51 ns maximum
t_{PHL}	6 ns minimum, 51 ns maximum

* 1.3 Module function. This module contains four 4-bit shift registers that have independent serial and parallel inputs, but common clock and mode control inputs.

1.3.1 Input to output relationship. Each shift register has the input to output relationship of table I for positive logic.

1.3.2 Parallel outputs Paralleling of outputs is not permissible

1.3.3 Operational description

1.3.3.1 Serial right-shift. The serial right-shift operation occurs when the mode control (M) is at low level; input data is applied to the serial input, and the clock transitions from high level to low level. Each clock transition shifts the output data to the next most significant position while loading the least significant position with the serial input data. Output QA is the least significant position of the right-shift operation.

1.3.3.2 Parallel operation The parallel operation occurs when the mode control (M) is at high level, the input data is applied to the inputs A, B, C, and D, and the clock transitions from high level to low level. Parallel transfer of data implies that the data present at the A, B, C, and D inputs prior to the clock transition are transferred to QA, QB, QC, and QD, respectively, after the clock transition

1.3.3.3 Changing the mode (M). Changes at the mode control input should normally be made when the clock is at low level to protect the register contents. However, a mode control change from low level to high level is permissible when the clock is at high level. Minimum mode control setup and hold times are 20 ns and are as defined on figure 24 for T_1, T_2, and T_3.

1.3.3.4 Serial left-shift. The serial left-shift operation occurs when the mode control (M) is at high level; data is present at the D input; QD output is connected to the C input; QC output is connected to the B input; QB output is connected to the A input; and the clock transitions from high level to low level. Output QD is the least significant position for the left-shift operation.

* 1.3.4 Parallel and serial setup time (T_S) and hold time (T_H). The logic level applied to a data input (SER, A, B, C, D) must be present at least 20 ns prior to the clock transition from high level to low level, but it is not necessary to maintain this level after the transition.

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TABLE I. Input to output relationship.

Mode control	Previous output state				Input state prior to transition of clock from high to low					Output after transition of clock from high to low			
M	A	B	C	D	A	B	C	D	SER	A	B	C	D
H	X	X	X	X	L	L	L	L	X	L	L	L	L
H	X	X	X	X	L	L	L	H	X	L	L	L	H
H	X	X	X	X	L	L	H	L	X	L	L	H	L
H	X	X	X	X	L	L	H	H	X	L	L	H	H
H	X	X	X	X	L	H	L	L	X	L	H	L	L
H	X	X	X	X	L	H	L	H	X	L	H	L	H
H	X	X	X	X	L	H	H	L	X	L	H	H	L
H	X	X	X	X	L	H	H	H	X	L	H	H	H
H	X	X	X	X	H	L	L	L	X	H	L	L	L
H	X	X	X	X	H	L	L	H	X	H	L	L	H
H	X	X	X	X	H	L	H	I	X	H	L	H	L
H	X	X	X	X	H	L	H	H	X	H	L	H	H
H	X	X	X	X	H	H	L	L	X	H	H	L	L
H	X	X	X	X	H	H	L	H	X	H	H	L	H
H	X	X	X	X	H	H	H	L	X	H	H	H	L
H	X	X	X	X	H	H	H	H	X	H	H	H	H
L	X	X	X	X	X	X	X	X	H	H	X	X	X
L	H	X	X	X	X	X	X	X	H	H	H	X	X
L	H	H	X	X	X	X	X	X	H	H	H	H	X
L	H	H	H	X	X	X	X	X	H	H	H	H	H
L	X	X	X	X	X	X	X	X	L	L	X	X	X
L	L	X	X	X	X	X	X	X	L	L	L	X	X
L	L	L	X	X	X	X	X	X	L	L	L	L	X
L	L	L	L	X	X	X	X	X	L	L	L	L	L
H	X	X	X	X	X	X	X	H	X	X	X	X	H
H	X	X	X	H	X	X	H	H	X	X	X	H	H
H	X	X	H	H	X	H	H	H	X	X	H	H	H
H	X	H	H	H	H	H	H	H	X	H	H	H	H

NOTES:
1. L = Low-level voltage.
2. H = High-level voltage.
3. X = Don't care.

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* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARD

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.33 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JDJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

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3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/33-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

* 3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

* 3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

* 3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
Capacitance meter	Item C05	1
Oscilloscope	Item D01	1
Oscilloscope	Item D05	1
Square wave generator	Item H01	1
Pulse generator	Item J01	2
DC power supply	Item P01	4
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope probe	Item Y01	2
Oscilloscope probe	Item Y02	1
Resistors		
10Ω ±5%, 1/4 W		1
51Ω ±5%, 1/2 W		3
316Ω ±1%, 1/4 W		18
511Ω ±1%, 1/4 W		1
877Ω ±1%, 1/4 W		1
Value as required for figure 10, note 4		16
Value as required for figure 10, note 5		16

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Capacitors	No required
100 pF ±20%	16
Value as required for figure 10, note 2	16
Value as required for figure 10, note 3	16
Diode	
1N4150, or equivalent	16
Miscellaneous	
Circuit, divide by 5	1
Switch, SPST	1
Switch, SPDT	1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through EE shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows. A, B, C where applicable, followed by the remaining tests in the sequence. If test conditions A or B are successively repeated, the time interval between execution of each of these test conditions should be a minimum of 10 seconds to preclude detrimental effects due to module heating.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through EE in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through Y, and CC through EE, shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column, and test condition CC, shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through Y, and CC through EE shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests C through EE shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

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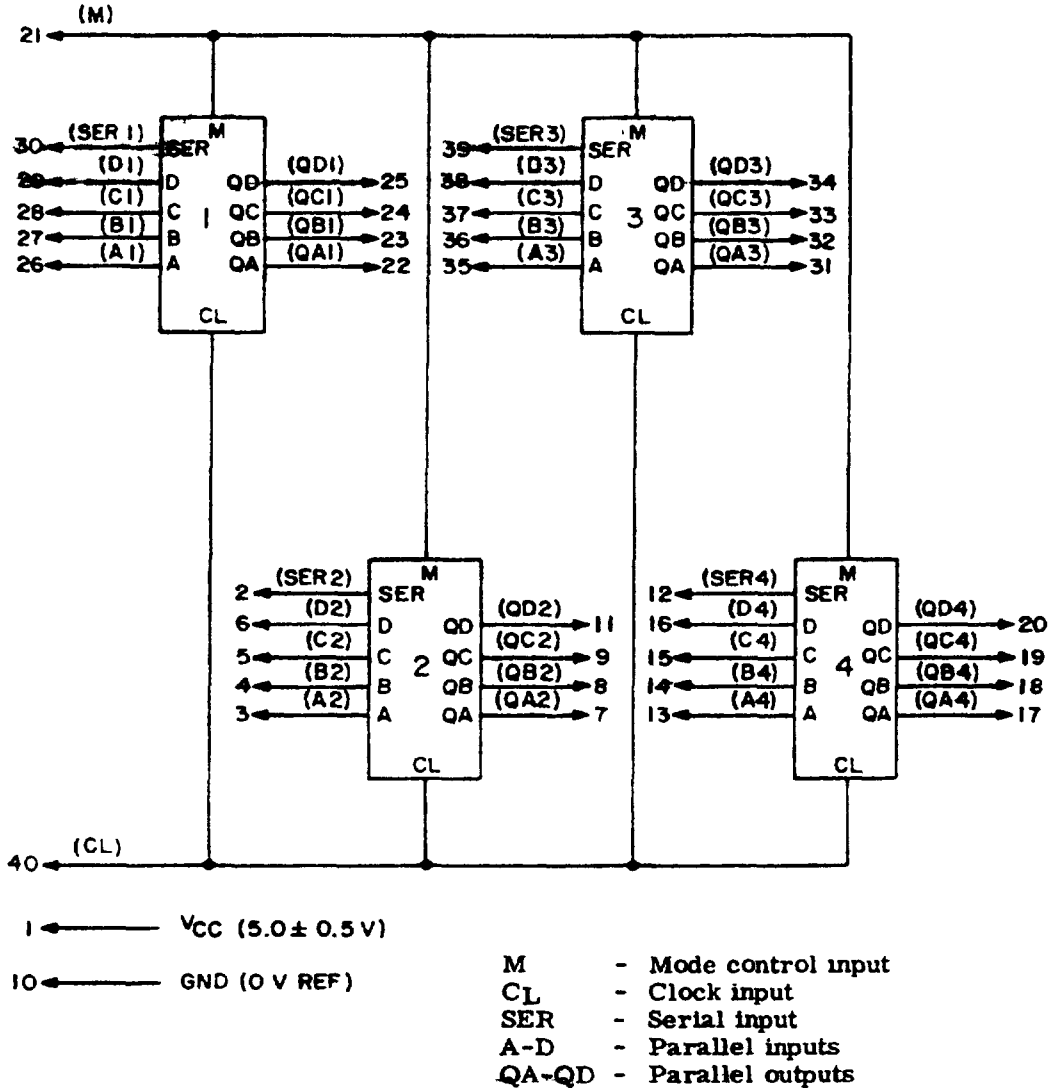


FIGURE 1. JDJ module function diagram.

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TABLE II. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_{OUT} (low-loaded)	III	S	0.01	0.80	0 01	0 80	0 01	0.80	0.01	0 80	V
B1-B24	Input breakdown current	IV	S	0.0	1.0	0 0	1 0	0 0	1.0	0 0	1 0	mA
C	Isolation	V	S	10	∞	1 0	∞	10	∞	1 0	∞	MΩ
D1-D20	Input capacitance (data)	VI	Q	2	25	2	25	2	25	2	25	pF
D21	Input capacitance (mode control)	VI	Q	2	50	2	50	2	50	2	50	pF
D22	Input capacitance (clock)	VI	Q	2	60	2	60	2	60	2	60	pF
E1-E20	Static output voltage, low level	VII	X	0.0	0.4	0 0	0.4	0.0	0.4	0.0	0.4	V
F1-F20	Static output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G4	Function, high level	IX	X	← See figure 9 →								
H1-H4	Function, low level	IX	X	← See figure 9 →								
J1-J20	Input current (data), high level	X	X	0	30	0	30	0	40	0	40	µA
J21-J23	Input current (mode control and clock), high level	X	X	0	240	0	240	0	320	0	320	µA
K1-K20	Input current (data), low level	XI	X	0.40	1.73	0 40	1 73	0 40	1.73	0 40	1.73	mA
K21-K23	Input current (mode control and clock), low level	XI	X	3.0	13.8	3 0	13 8	3 0	13.8	3 0	13.8	mA
L1-L16	Output short circuit current	XII	X	18	57	18	57	18	57	18	57	mA
M1,M2	Power supply current	XIII	S	1	288	1	288	1	288	1	288	mA
N1-N16	Transition time (t_{TLH})	XIV	S	4	45	4	45	4	50	4	50	ns
P1-P16	Transition time (t_{THL})	XIV	S	4	20	4	20	4	25	4	25	ns
R1-R16	Propagation delay time (t_{PHL})	XIV	S	6	46	6	46	6	51	6	51	ns

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S1-S16	Propagation delay time (t_{PLH})	XIV	S	6	46	6	46	6	51	6	51	ns
T1-T4	Maximum shift frequency:	XV	S									
	TA			112	170	112	170	112	170	112	170	ns
	TB			38	90	38	90	38	90	38	90	ns
	TC			120	136	120	136	120	136	120	136	ns
* U1-U32	Parallel setup time·	XVI	S									
	TA			21	72	21	72	21	72	21	72	ns
	TB			416	584	416	584	416	584	416	584	ns
	TC			900	1,100	900	1,100	900	1,100	900	1,100	ns
* V1-V32	Parallel hold time:	XVII	S									
	TA			450	603	450	603	450	603	450	603	ns
	TB			416	584	416	584	416	584	416	584	ns
	TC			900	1,100	900	1,100	900	1,100	900	1,100	ns
* W1-W8	Serial setup time:	XVIII	S									
	TA			471	622	471	622	471	622	471	622	ns
	TB			416	584	416	584	416	584	416	584	ns
	TC			900	1,100	900	1,100	900	1,100	900	1,100	ns
* X1-X8	Serial hold time	XIX	S									
	TA			0	53	0	53	0	53	0	53	ns
	TB			416	584	416	584	416	584	416	584	ns
	TC			900	1,100	900	1,100	900	1,100	900	1,100	ns
Y1-Y4	Mode control setup time:	XX	S									
	TA			210	283	210	283	210	283	210	283	ns
	TB			118	202	118	202	118	202	118	202	ns
	TC			380	420	380	420	380	420	380	420	ns
Z1-Z16	Output crosstalk, low level	XXI	Q	0	800	0	800	0	800	0	800	mV
Z17-Z32	Output crosstalk, duration	XXI	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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TABLE II. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
AA1-AA16	Output crosstalk, high level	XXII	Q	0	500	0	500	0	500	0	500	mV
BB	Power supply transient amplitude	XXIII	Q	0	400	0	400	0	400	0	400	mV
CC	Power supply transient rise time	XXIV	S	1.0	2.0	1.0	2.0	1.0	2 0	1.0	2 0	ms
DD1-DD22	Input clamp diode voltage	XXV	Q	-0.4	-1.5	-0.4	-1.5	-0 4	-1.5	-0 4	-1 5	V
EE1-EE28	False shift immunity	XXVI	X	← See table XXVI →								

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only.

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5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/33-1 supersedes part number M28787/33. The two parts are interchangeable.

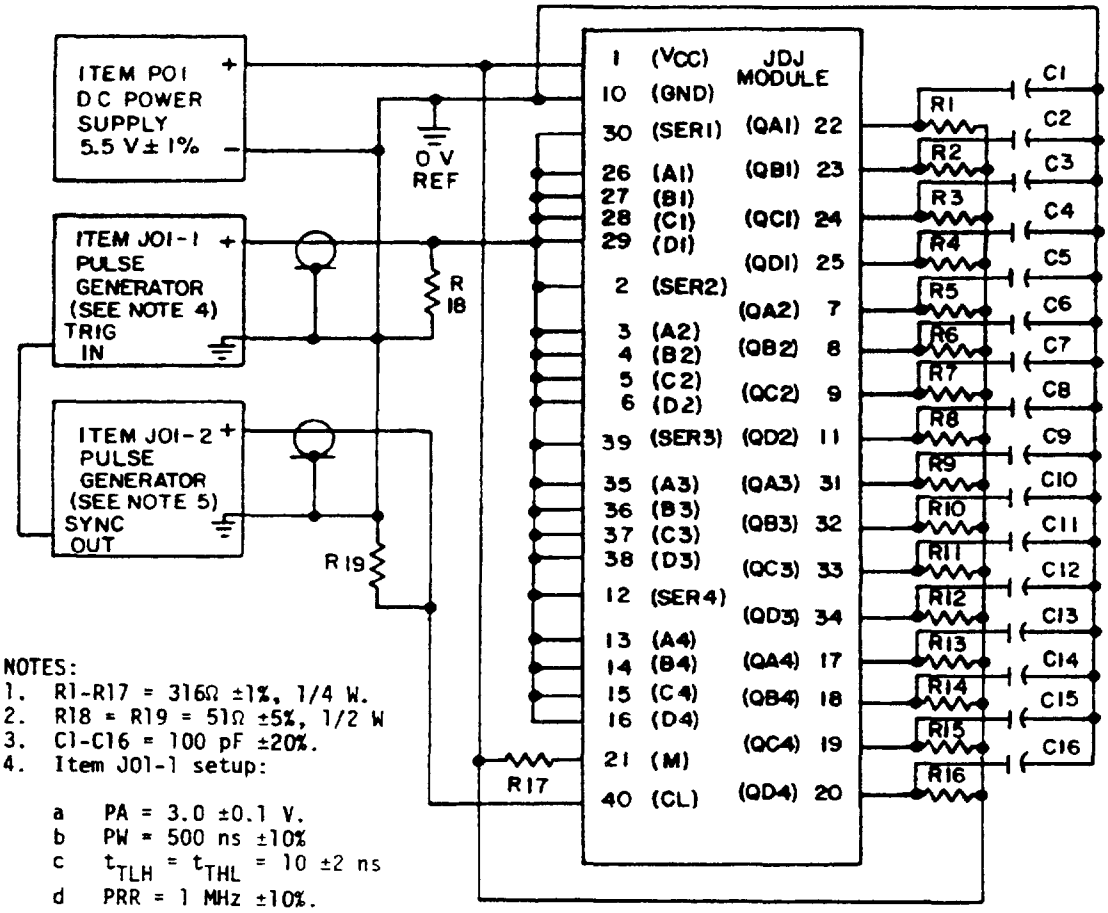
6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. R1-R17 = 316Ω ±1%, 1/4 W.
2. R18 = R19 = 51Ω ±5%, 1/2 W
3. C1-C16 = 100 pF ±20%.
4. Item J01-1 setup:

   a PA = 3.0 ±0.1 V.
   b PW = 500 ns ±10%
   c $t_{TLH} = t_{THL}$ = 10 ±2 ns
   d PRR = 1 MHz ±10%.

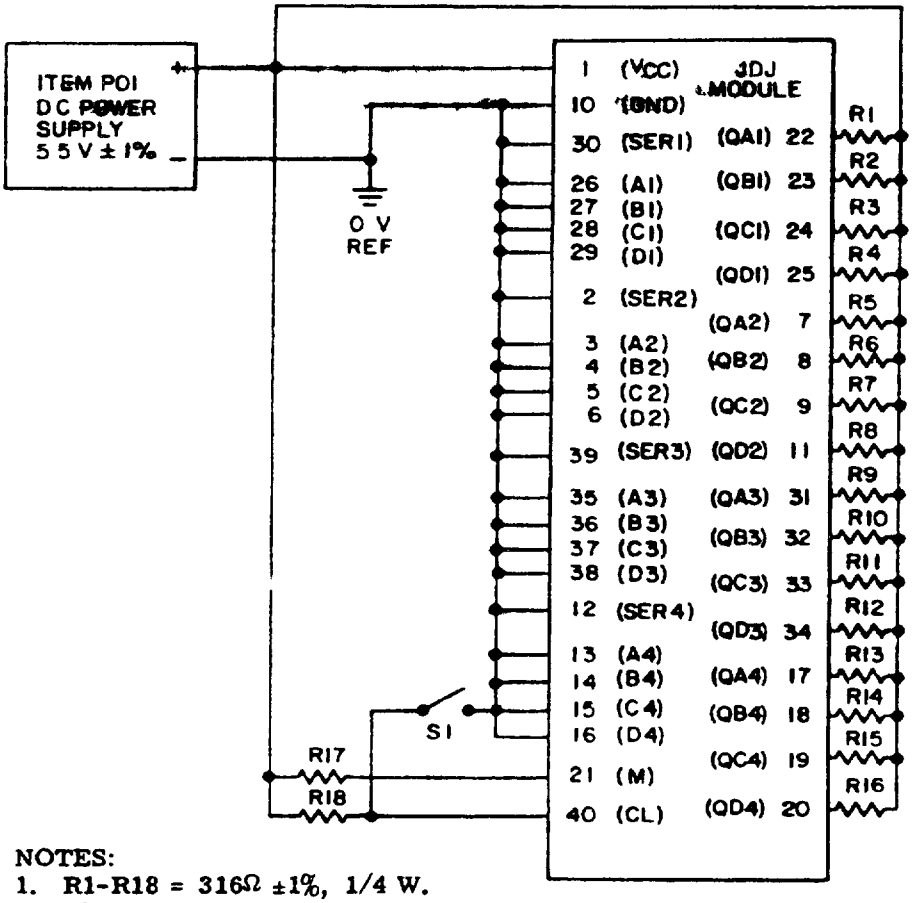
5 Item J01-2 setup:

   a. PA = 3.0 ±0.1 V.
   b. PW = 250 ns ±10%.
   c. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   d. PRR = 2 MHz ±10%.

6. $T_S(1) = T_S(0)$ = 375 ns ±10%. (see figure 15).
7. Conductor lengths should be as short as possible.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R18 = 316Ω ±1%, 1/4 W.
2. S1 = SPST.
3. After making all circuit connections, close switch S1.

FIGURE 3. Component temperature test circuit.

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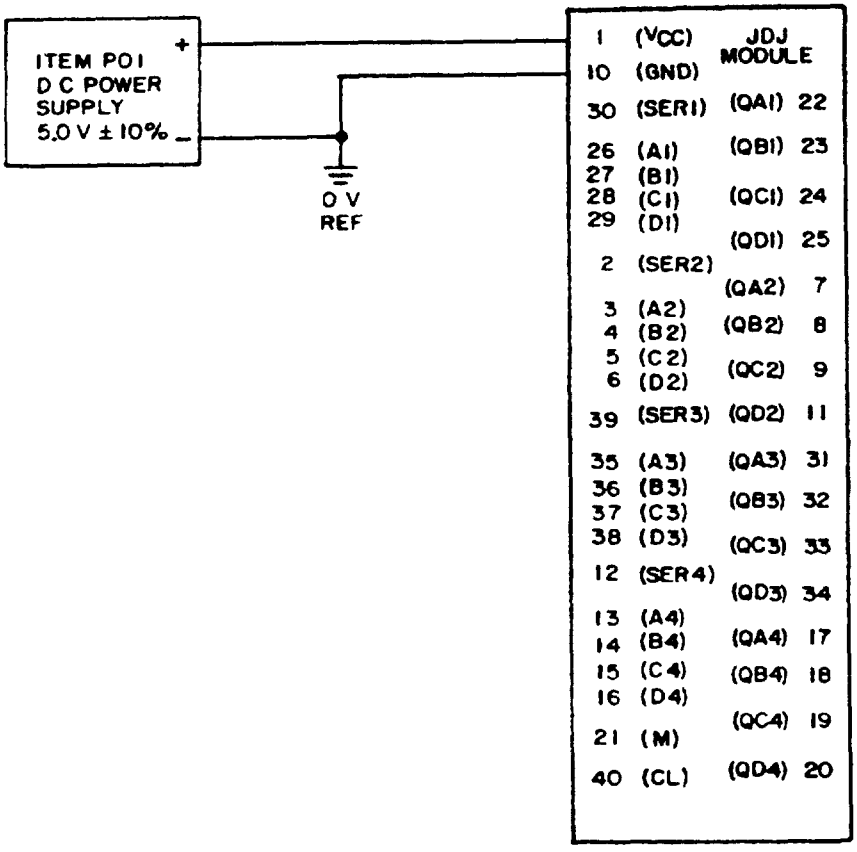


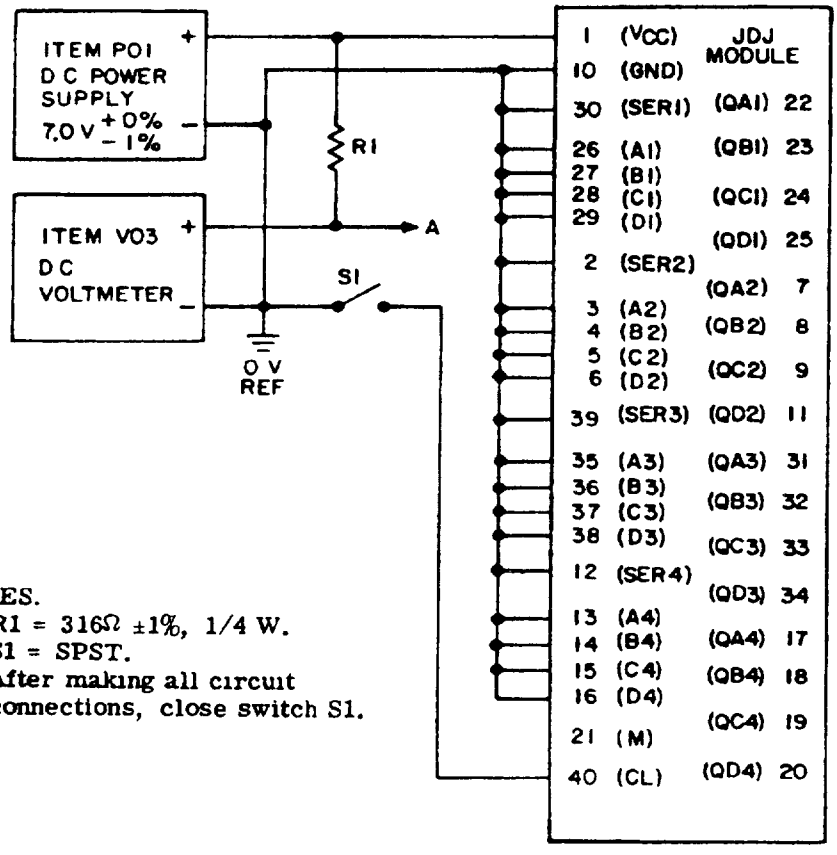
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A1	QA1 V_{OUT} (low-loaded)	22	0.01	0.80	V	5
A2	QB1	23				
A3	QC1	24				
A4	QD1	25				
A5	QA2	7				
A6	QB2	8				
A7	QC2	9				
A8	QD2	11				
A9	QA3	31				
A10	QB3	32				
A11	QC3	33				
A12	QD3	34				
A13	QA4	17				
A14	QB4	18				
A15	QC4	19				
A16	QD4 V_{OUT} (low-loaded)	20	0.01	0.80	V	5

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NOTES.
1. R1 = 316Ω ±1%, 1/4 W.
2. S1 = SPST.
3. After making all circuit connections, close switch S1.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdon current test.

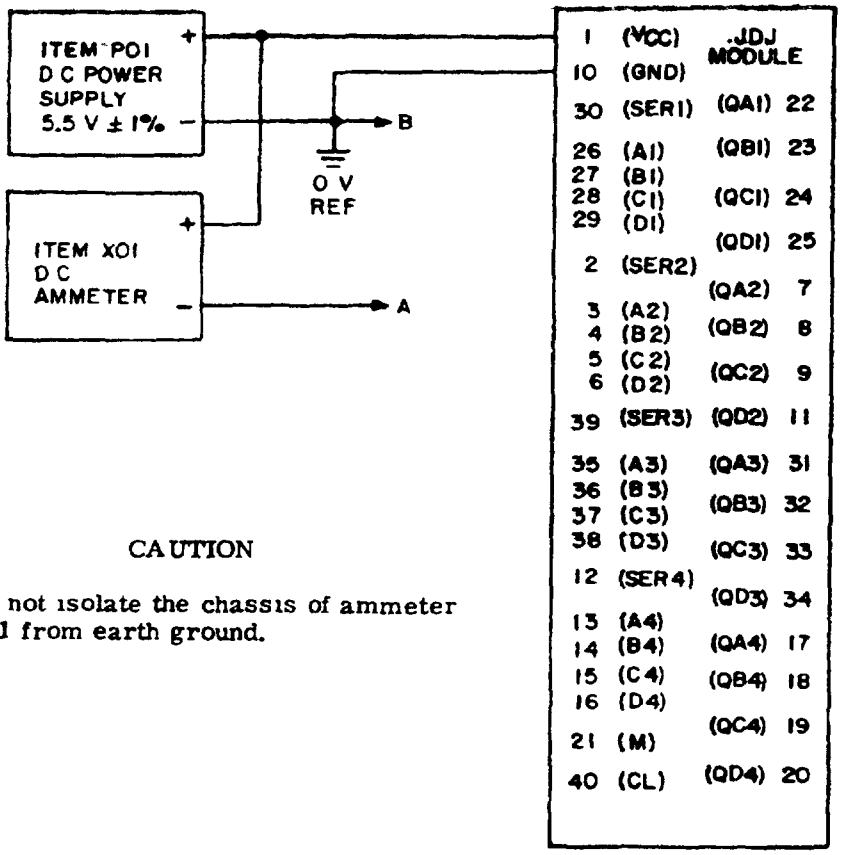
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A1 Input breakdown current	26	21	0 0	1.0	mA	6
B2	B1	27					
B3	C1	28					
B4	D1	29					
B5	A2	3					
B6	B2	4					
B7	C2	5					
B8	D2	6					
B9	A3	35					
B10	B3	36					
B11	C3	37					
B12	D3	38					
B13	A4	13					
B14	B4	14					
B15	C4	15					
B16	D4	16					
B17	CL	40	21				
B18	M	21	40				
B19	M Input breakdown current	21	Open	0.0	1.0	mA	6

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TABLE IV. Input breakdown current test - Continued

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B20	CL	Input breakdown current	40	Open	0.0	1.0	mA	6
B21	SER1	↓	30	↓	↓	↓	↓	↓
B22	SER2	↓	2	↓	↓	↓	↓	↓
B23	SER3	↓	39	↓	↓	↓	↓	↓
B24	SER4	Input breakdown current	12	Open	0.0	1.0	mA	6

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

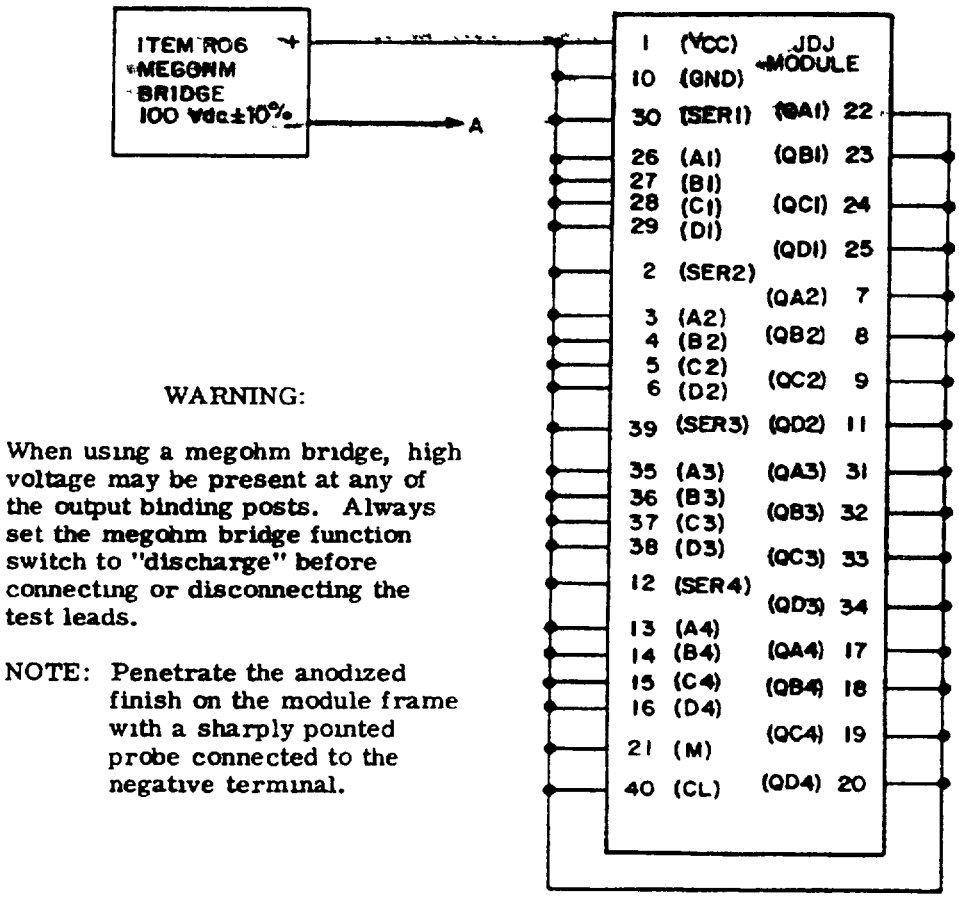
FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Isolation test

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
C	Isolation	Frame	10	∞	MΩ	7

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 7. Isolation test circuit for table V.

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TABLE VI. Input capacitance test.

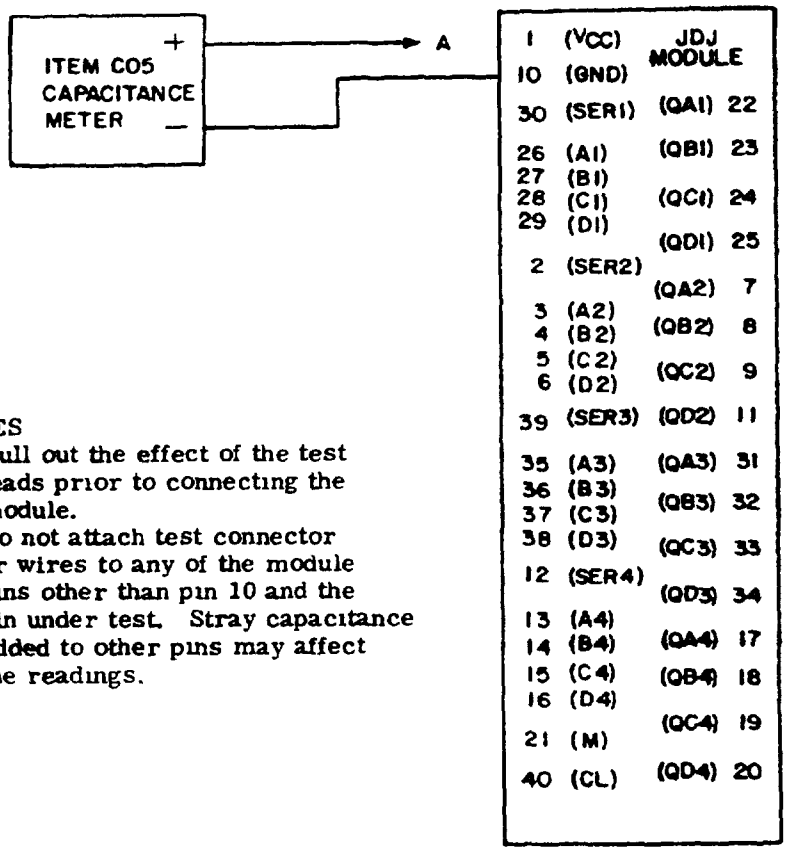
Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D1	SER1 Input capacitance	30	2	25	pF	8
D2	A1	26				
D3	B1	27				
D4	C1	28				
D5	D1	29				
D6	SER2	2				
D7	A2	3				
D8	B2	4				
D9	C2	5				
D10	D2	6				
D11	SER3	39				
D12	A3	35				
D13	B3	36				
D14	C3	37				
D15	D3	38				
D16	SER4	12				
D17	A4	13				
D18	B4	14				
D19	C4 Input capacitance	15	2	25	pF	8

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TABLE VI. Input capacitance test - Continued

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
D20	D4 Input capacitance	16	2	25	pF	8
D21	M Input capacitance	21	2	50	pF	8
D22	CL Input capacitance	40	2	60	pF	8

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NOTES
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 8. Input capacitance test circuit for table VI.

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TABLE VII. Static output voltage (low-level) test.

Test condition	Parameter		Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	QA1	Static output voltage, low level	2,12,21, 30,39	26,27,28, 29	22	Open	40	0.0	0.4	V	9
E2	QB1	↓	↓	↓	23	↓	↓	↓	↓	↓	↓
E3	QC1	↓	↓	↓	24	↓	↓	↓	↓	↓	↓
E4	QD1	↓	↓	26,27,28, 29	25	↓	↓	↓	↓	↓	↓
E5	QA2	↓	↓	3,4,5,6	7	↓	↓	↓	↓	↓	↓
E6	QB2	↓	↓	↓	8	↓	↓	↓	↓	↓	↓
E7	QC2	↓	↓	↓	9	↓	↓	↓	↓	↓	↓
E8	QD2	↓	↓	3,4,5,6	11	↓	↓	↓	↓	↓	↓
E9	QA3	↓	↓	35,36,37, 38	31	↓	↓	↓	↓	↓	↓
E10	QB3	↓	↓	↓	32	↓	↓	↓	↓	↓	↓
E11	QC3	↓	↓	↓	33	↓	↓	↓	↓	↓	↓
E12	QD3	↓	↓	35,36,37, 38	34	↓	↓	↓	↓	↓	↓
E13	QA4	↓	↓	13,14,15, 16	17	↓	↓	↓	↓	↓	↓
E14	QB4	↓	↓	↓	18	↓	↓	↓	↓	↓	↓
E15	QC4	↓	↓	↓	19	↓	↓	↓	↓	↓	↓
E16	QD4	Static output voltage, low level	2,12,21, 30,39	13,14,15, 16	20	Open	40	0.0	0.4	V	9

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Check the source to verify that this is the current version before use.

TABLE VII. Static output voltage (low-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E17	QA1 Static output voltage, low level	26,27,28, 29	2,12,21, 30,39	22	Open	40	0.0	0.4	V	9
E18	QA2	3,4,5,6		7						
E19	QA3	35,36,37, 38		31						
E20	QA4 Static output voltage, low level	13,14,15, 16	2,12,21, 30,39	17	Open	40	0.0	0.4	V	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:36Z
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TABLE VIII. Static output voltage (high-level) test.

Test condition	Parameter		Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	QA1	Static output voltage, high level	3,4,5,6, 13,14,15, 16,21,26, 27,28,29, 35,36,37, 38	2,12,30, 39	22	Open	40	2.4	4.5	V	9
F2	QB1				23						
F3	QC1				24						
F4	QD1		3,4,5,6, 13,14,15, 16,21,26, 27,28,29, 35,36,37, 38		25						
F5	QA2		3,4,5,6, 21		7						
F6	QB2				8						
F7	QC2				9						
F8	QD2		3,4,5,6, 21		11						
F9	QA3		21,35,36, 37,38		31						
F10	QB3				32						
F11	QC3				33						
F12	QD3	Static output voltage, high level	21,35,36, 37,38	2,12,30, 39	34	Open	40	2.4	4.5	V	9

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TABLE VIII. Static output voltage (high-level) test - Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F13	QA4 Static output voltage, high level	13,14,15, 16,21	2,12,30, 39	17	Open	40	2.4	4.5	V	9
F14	QB4	↓	↓	18	↓	↓	↓	↓	↓	↓
F15	QC4	↓	↓	19	↓	↓	↓	↓	↓	↓
F16	QD4	13,14,15, 16,21	2,12,30, 39	20	↓	↓	↓	↓	↓	↓
F17	QA1	2,12,30, 39	21,26,27, 28,29	22	↓	↓	↓	↓	↓	↓
F18	QA2	↓	3,4,5,6, 21	7	↓	↓	↓	↓	↓	↓
F19	QA3	↓	21,35,36, 37,38	31	↓	↓	↓	↓	↓	↓
F20	QA4 Static output voltage, high level	2,12,30, 39	13,14,15, 16,21	17	Open	40	2 4	4.5	V	9

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TABLE IX. Function (high-level and low-level) test

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
G1	Function, high level	2,12,30, 39	26,27,28, 29	Open	40	21	See figure 9			9
G2			3,4,5,6							
G3			35,36,37, 38							
G4	Function, high level	2,12,30, 39	13,14,15, 16							
H1	Function, low level	26,27,28, 29	2,12,30, 39							
H2		3,4,5,6								
H3		35,36,37, 38								
H4	Function, low level	13,14,15, 16	2,12,30, 39	Open	40	21	See figure 9			9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:36Z
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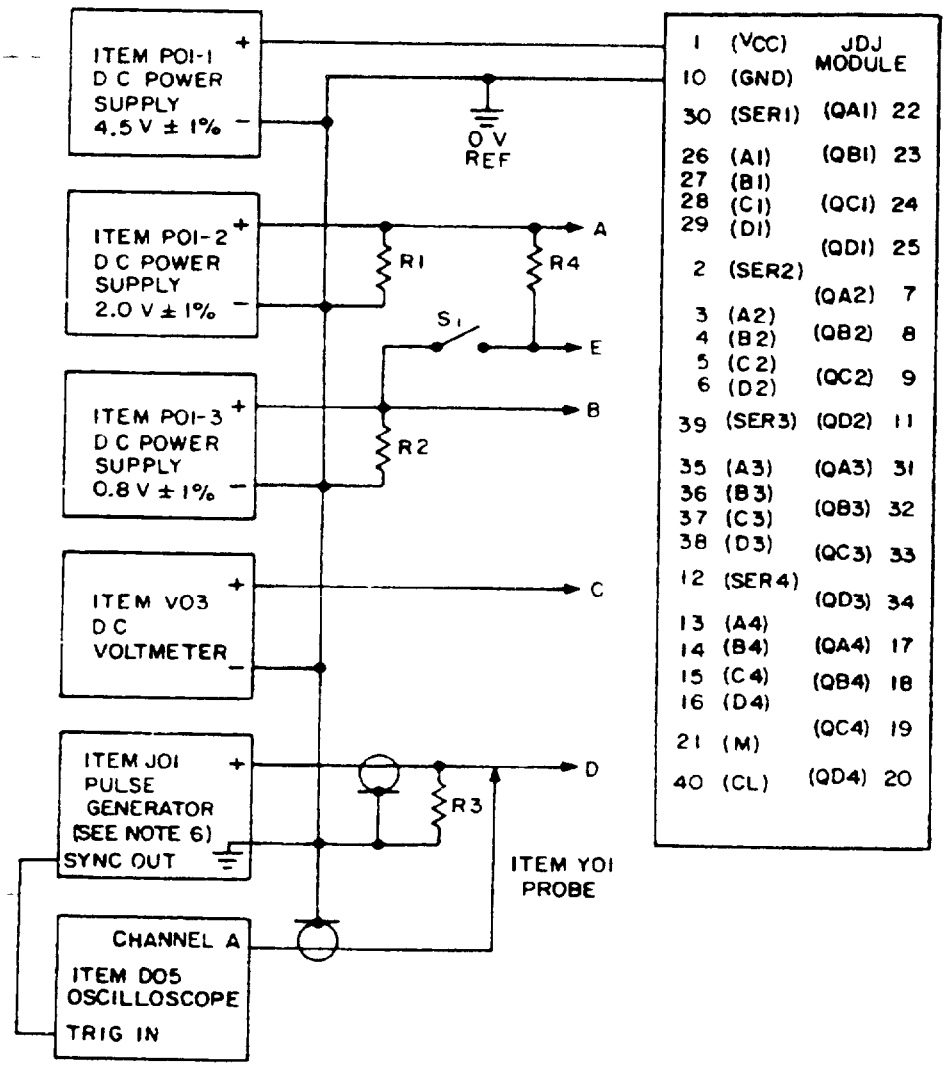


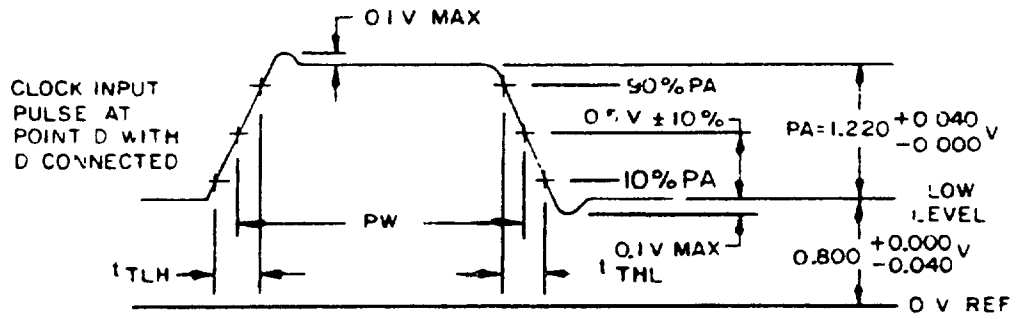
FIGURE 9 Static output voltage (low-level and high-level) and function (high-level and low-level) test circuit for tables VII, VIII, and IX.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W
2. R4 = 316Ω ±1%, 1/4 W.
3. S1 = SPST
4. Load all outputs of each shift register circuit under test with a load as specified on figure 10
5. Test procedure for each condition of test conditions E and F:

   a. Open switch S1.
   b. Make all connections for a test condition as specified.
   c. Close switch S1.
   d. Take measurement.

6. Item J01 setup:



   a. PW = 500 ns ±10%.
   b. t_{TLH} = t_{THL} = 25 ns ±10%.
   c. PRR = 30 kHz ±50%.
   d. Put item J01 in single-pulse mode.
   e. Using item V03, adjust the dc logic low level.

7. For test conditions G and H,

   L = 0.0 V minimum, 0.4 V maximum.
   H = 2.4 V minimum, 4.5 V maximum.

FIGURE 9. Static output voltage (low-level and high-level) and function (high-level and low-level) test circuit for tables VII, VIII, and IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:36Z
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8. Test procedure for each condition of test conditions G and H:

a. Open switch S1.
b. Make all connections for a test condition as specified.
c Apply one pulse to the circuit under test.
d. Verify that the output states are as follows:

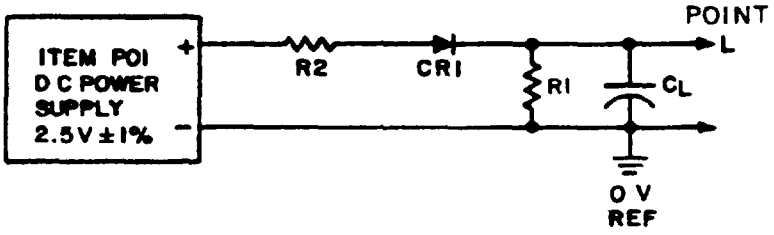
Test condition G				Test condition H			
QA	QB	QC	QD	QA	QB	QC	QD
L	L	L	L	H	H	H	H

The output logic states for each shift register circuit under test must be detected simultaneously.
e. Close switch S1.
f. Apply four pulses to the circuit under test and after each pulse is applied, verify that the output states are as follows:

Pulse No.	Test condition G				Test condition H			
	QA	QB	QC	QD	QA	QB	QC	QD
1	H	L	L	L	L	H	H	H
2	H	H	L	L	L	L	H	H
3	H	H	H	L	L	L	L	H
4	H	H	H	H	L	L	L	L

FIGURE 9. Static output voltage (low-level and high-level) and function (high-level and low-level) test circuit for tables VII, VIII, and IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:36Z
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NOTES
1. CR1 = 1N4150, or equivalent.
2. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions N, P, R, S, and T.
3. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions Z and AA.
4. R1 = Select value for current flow of 800 µA ±1% into point L when L is held at 2 4 Vdc (3.01 kΩ ±1%).
5. R2 = Select value for current flow of 17.3 mA ±1% from point L to ground
6. C_L is not required for dc electrical test conditions E, F, G, and H
7. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76 2 mm)
8. During output switching, item PO1 shall have a maximum overshoot of 500 mV

* FIGURE 10. JDJ logic load circuit

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TABLE X. Input current (high-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	SER1	Input current, high level	30	Open	Open	0	30	µA	11
J2	SER2		2						
J3	SER3		39						
J4	SER4		12		Open				
J5	A1		26		21,27, 28,29				
J6	B1		27		21,26, 28,29				
J7	C1		28		21,26, 27,29				
J8	D1		29		21,26, 27,28				
J9	A2		3		4,5,6,21				
J10	B2		4		3,5,6,21				
J11	C2		5		3,4,6,21				
J12	D2		6		3,4,5,21				
J13	A3		35		21,36, 37,38				
J14	B3		36		21,35, 37,38				
J15	C3	Input current, high level	37	Open	21,35, 36,38	0	30	µA	11

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J16	D3 Input current, high level	38	Open	21,35, 36,37	0	30	μA	11
J17	A4	13		14,15, 16,21				
J18	B4	14		13,15, 16,21				
J19	C4	15		13,14, 16,21				
J20	D4	16		13,14, 15,21		30		
J21	CL	40		Open		240		
J22	CL	40		21		240		
J23	M Input current, high level	21	Open	40	0	240	μA	11

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TABLE XI. Input current (low-level) test.

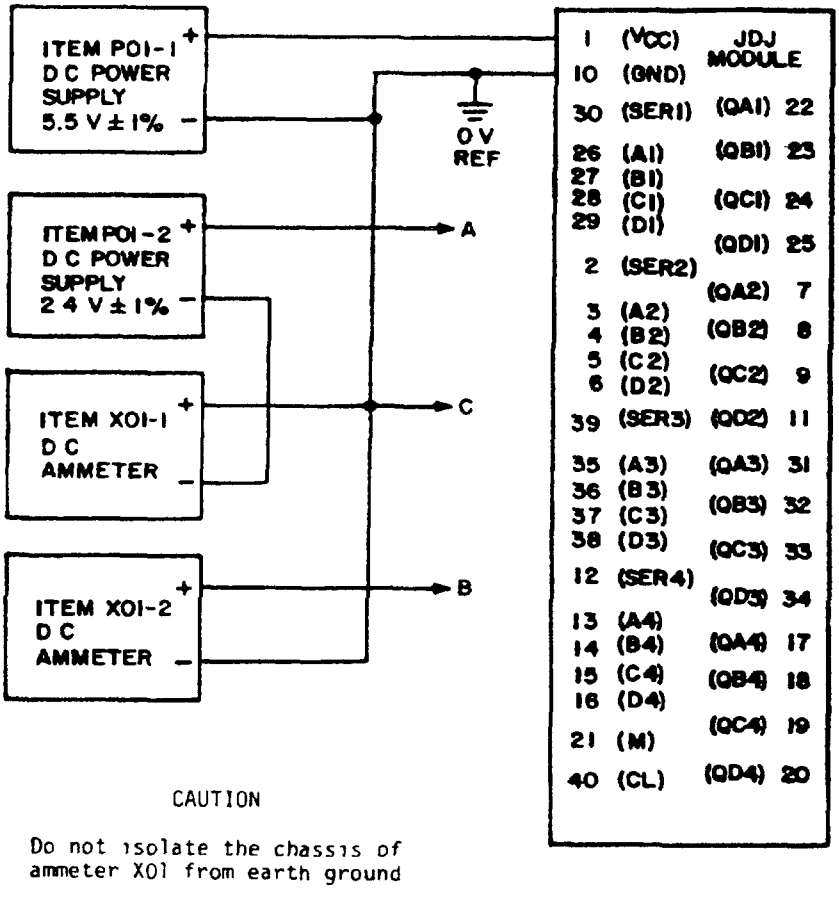
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
K1	SER1	Input current, low level	Open	30	21	0.40	1.73	mA	11
K2	SER2			2					
K3	SER3			39					
K4	SER4			12	21				
K5	A1			26	Open				
K6	B1			27					
K7	C1			28					
K8	D1			29					
K9	A2			3					
K10	B2			4					
K11	C2			5					
K12	D2			6					
K13	A3			35					
K14	B3			36					
K15	C3			37					
K16	D3			38					
K17	A4			13					
K18	B4			14					
K19	C4	Input current, low level	Open	15	Open	0.40	1.73	mA	11

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K20	D4 Input current, low level	Open	16	Open	0.40	1.73	mA	11
K21	CL		40	21	3.0	13.8		
K22	CL		40	Open	3.0	13.8		
K23	M Input current, low level	Open	21	Open	3.0	13.8	mA	11

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTES
1 Since the ammeters are not used simultaneously, only one dc ammeter X01 is required
2 With point A disconnected, ammeter X01-1 should indicate zero current on the 1 µA range

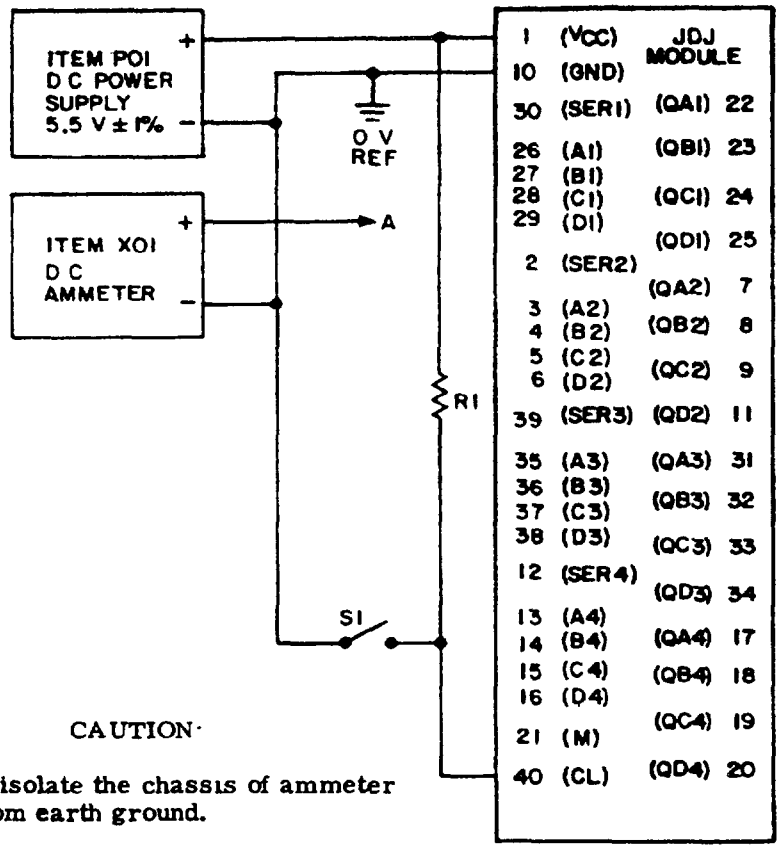
FIGURE 11 Input current (high-level and low-level) test circuit for tables X and XI

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TABLE XII Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
L1	QA1 Output short circuit current	22	18	57	mA	12
L2	QB1	23				
L3	QC1	24				
L4	QD1	25				
L5	QA2	7				
L6	QB2	8				
L7	QC2	9				
L8	QD2	11				
L9	QA3	31				
L10	QB3	32				
L11	QC3	33				
L12	QD3	34				
L13	QA4	17				
L14	QB4	18				
L15	QC4	19				
L16	QD4 Output short circuit current	20	18	57	mA	12

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 316Ω ±1%, 1/4 W.
2. S1 = SPST.
3. Connect the module and test equipments as shown, then close switch S1.
4. Only one output shall be shorted at a time and the measurement taken within 5 seconds.

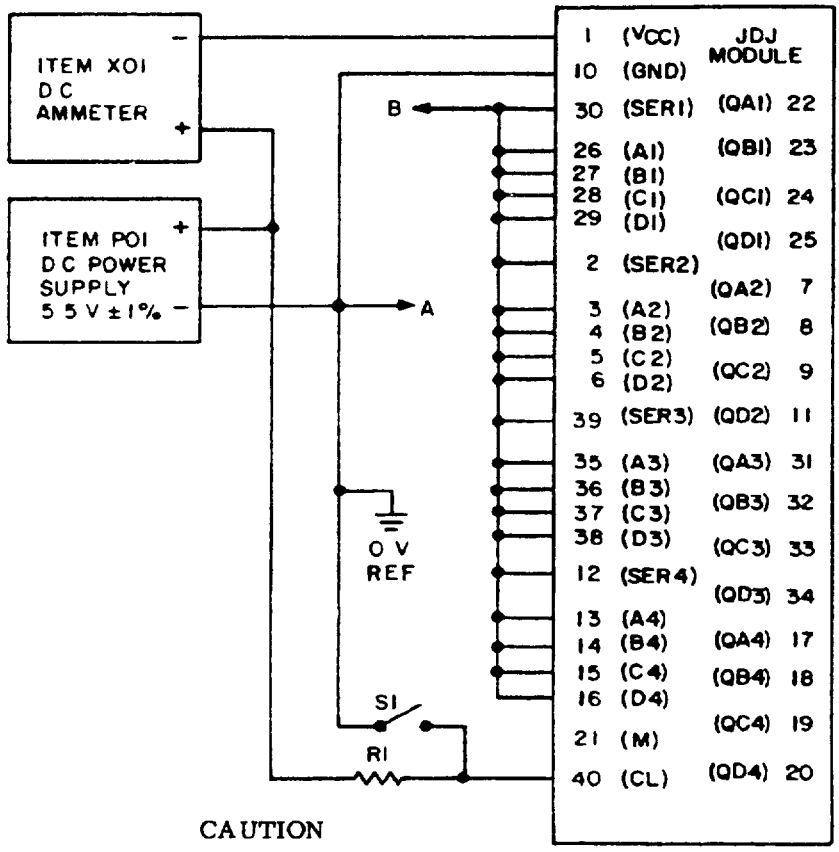
FIGURE 12. Output short circuit current test circuit for table XII.

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TABLE XIII. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, high level	Open	Open	1	288	mA	13
M2	Power supply current, low level	B	A	1	288	mA	13

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES
1. R1 = 316Ω ±1%, 1/4 W.
2. S1 = SPST.
3. Connect the module and test equipments as shown, then close switch S1.
4. Open switch S1.
5. Close switch S1.

FIGURE 13. Power supply current (high-level and low-level) test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N1	QA1	Transition time (t_{TLH})	26,27,28,29	22	4	45	ns	14,15
N2	QB1			23				
N3	QC1			24				
N4	QD1		26,27,28,29	25				
N5	QA2		3,4,5,6	7				
N6	QB2			8				
N7	QC2			9				
N8	CD2		3,4,5,6	11				
N9	QA3		35,36,37,38	31				
N10	QB3			32				
N11	QC3			33				
N12	QD3		35,36,37,38	34				
N13	QA4		13,14,15,16	17				
N14	QB4			18				
N15	QC4			19				
N16	QD4	Transition time (t_{TLH})	13,14,15,16	20	4	45	ns	14,15

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	QA1	Transition time (t_{THL})	26,27,28,29	22	4	20	ns	14,15
P2	QB1			23				
P3	QC1			24				
P4	QD1		26,27,28,29	25				
P5	QA2		3,4,5,6	7				
P6	QB2			8				
P7	QC2			9				
P8	QD2		3,4,5,6	11				
P9	QA3		35,36,37,38	31				
P10	QB3			32				
P11	QC3			33				
P12	QD3		35,36,37,38	34				
P13	QA4		13,14,15,16	17				
P14	QB4			18				
P15	QC4			19				
P16	QD4	Transition time (t_{THL})	13,14,15,16	20	4	20	ns	14,15

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

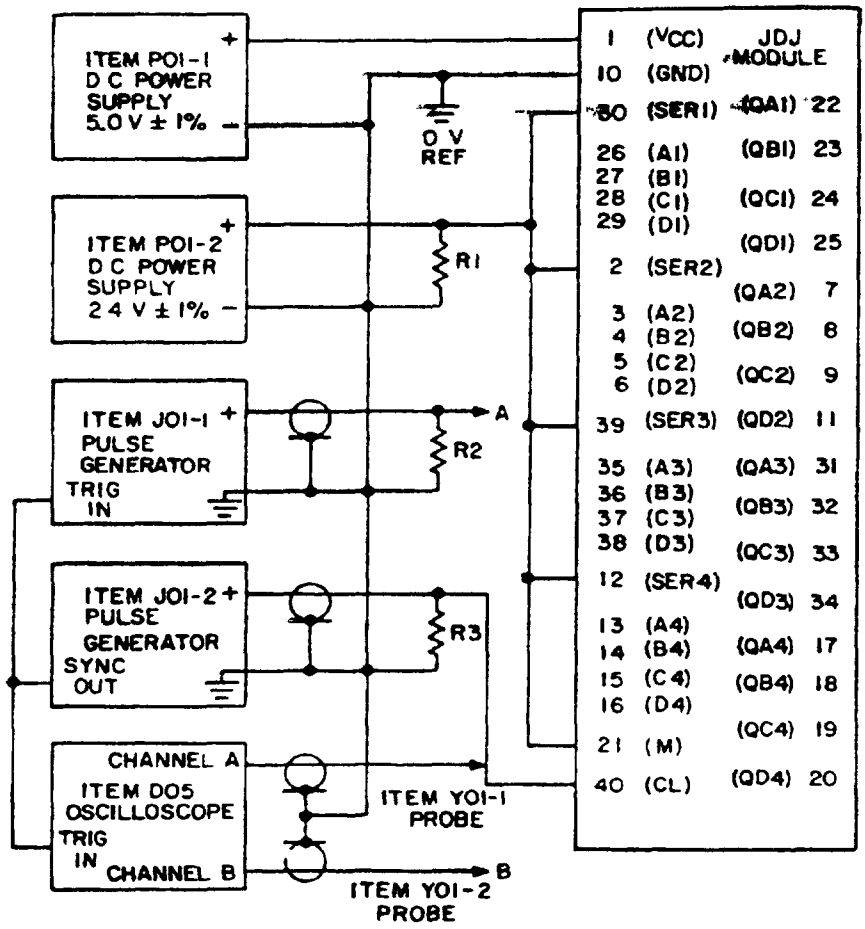
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
R1	CL to QA1	Propagation delay time (t_{PHL}), C_L = 100 pF	26,27,28,29	22	6	46	ns	14,15
R2	CL to QB1			23				
R3	CL to QC1			24				
R4	CL to QD1		26,27,28,29	25				
R5	CL to QA2		3,4,5,6	7				
R6	CL to QB2			8				
R7	CL to QC2			9				
R8	CL to QD2		3,4,5,6	11				
R9	CL to QA3		35,36,37,38	31				
R10	CL to QB3			32				
R11	CL to QC3			33				
R12	CL to QD3		35,36,37,38	34				
R13	CL to QA4		13,14,15,16	17				
R14	CL to QB4			18				
R15	CL to QC4			19				
R16	CL to QD4	Propagation delay time (t_{PHL}), C_L = 100 pF	13,14,15,16	20	6	46	ns	14,15

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TABLE XIV Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
S1	CL to QA1	Propagation delay time (t_{PLH}), C_L = 100 pF	26,27,28,29	22	6	46	ns	14,15
S2	CL to QB1			23				
S3	CL to QC1			24				
S4	CL to QD1		26,27,28,29	25				
S5	CL to QA2		3,4,5,6	7				
S6	CL to QB2			8				
S7	CL to QC2			9				
S8	CL to QD2		3,4,5,6	11				
S9	CL to QA3		35,36,37,38	31				
S10	CL to QB3			32				
S11	CL to QC3			33				
S12	CL to QD3		35,36,37,38	34				
S13	CL to QA4		13,14,15,16	17				
S14	CL to QB4			18				
S15	CL to QC4			19				
S16	CL to QD4	Propagation delay time (t_{PLH}), C_L = 100 pF	13,14,15,16	20	6	46	ns	14,15

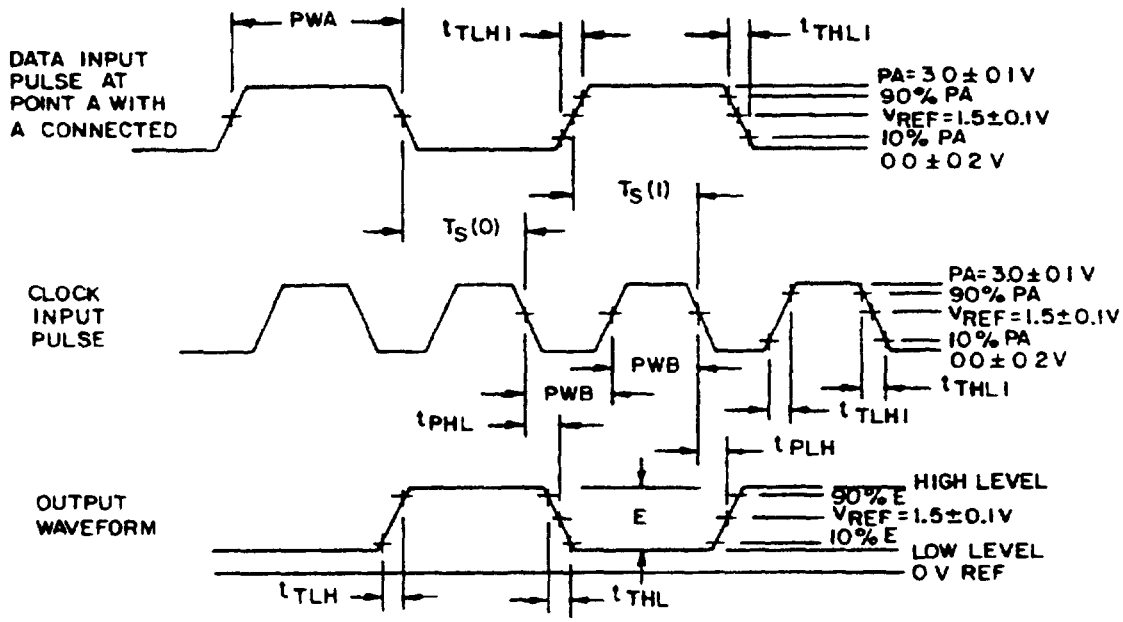
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load only the output under test with a load as specified on figure 10.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:

   a. PWA = 500 ns ±10%.
   b PWB = 250 ns ±10%.
   c. $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   d. $T_S(1) = T_S(0) = 375$ ns ±10%.

2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

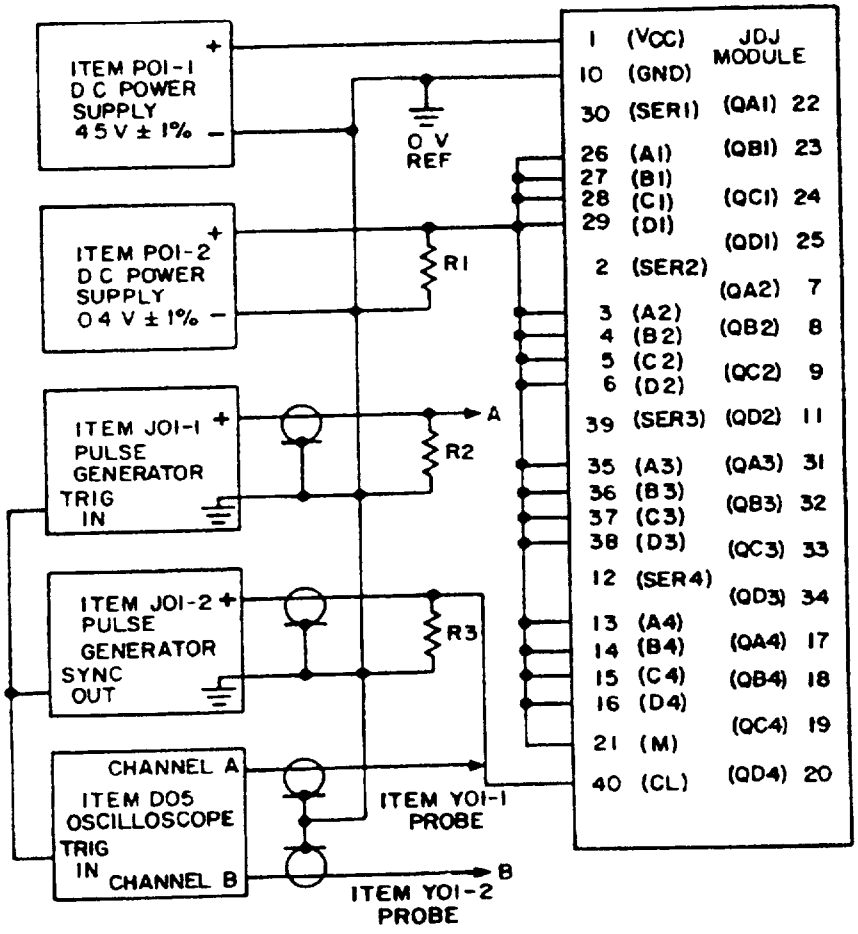
* FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV Maximum shift frequency test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient						Unit	Test figure
		A to pin	B to pin	TA		TB		TC			
				Min	Max	Min	Max	Min	Max		
T1	Maximum shift frequency	30	25	112	170	38	90	120	136	ns	16,17
T2	↓	2	11	↓	↓	↓	↓	↓	↓	↓	↓
T3	↓	39	34	↓	↓	↓	↓	↓	↓	↓	↓
T4	Maximum shift frequency	12	20	112	170	38	90	120	136	ns	16,17

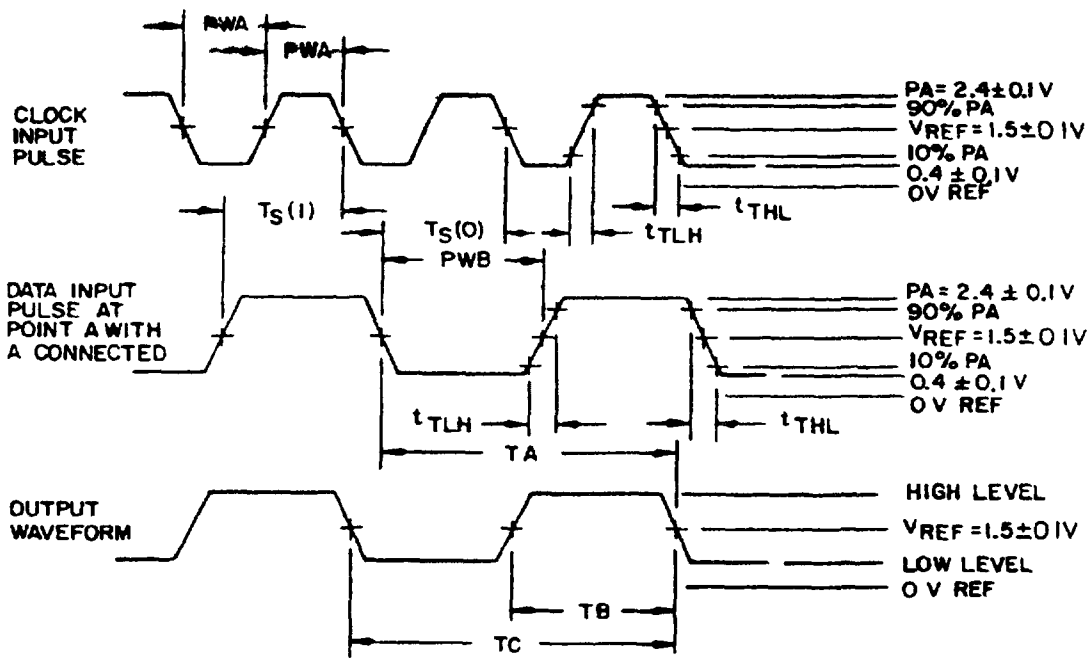
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NOTES:
1. R1 = 10Ω ±5%, 1/4 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. Load all outputs with a load as specified on figure 10.

FIGURE 16. Maximum shift frequency test circuit for table XV.

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NOTES
1. Input pulse parameters:

   a. PWA = 32 ±2 ns.
   b. PWB = 64 ±4 ns.
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   d. $T_S(1) = T_S(0) = 48$ ns ±5%.

2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

* FIGURE 17. Maximum shift frequency test circuit waveforms for table XV.

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TABLE XVI. Parallel setup time (in phase, T_S (0) and T_S (1)) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient						Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
							Min	Max	Min	Max	Min	Max		
U1	QA1	Parallel setup time, in phase, T_S (0)	2,12,21, 30,39	26,27,28,29	Open	22	21	72	416	584	900	1,100	ns	18,19
U2	QB1					23								
U3	QC1					24								
U4	QD1			26,27,28,29		25								
U5	QA2			3,4,5,6		7								
U6	QB2					8								
U7	QC2					9								
U8	QD2			3,4,5,6		11								
U9	QA3			35,36,37,38		31								
U10	QB3					32								
U11	QC3					33								
U12	QD3			35,36,37,38		34								
U13	QA4			13,14,15,16		17								
U14	QB4					18								
U15	QC4					19								
U16	QD4	Parallel setup time, in phase, T_S (0)	2,12,21, 30,39	13,14,15,16	Open	20	21	72	416	584	900	1,100	ns	18,19

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TABLE XVI. Parallel setup time (in phase, T_S (0) and T_S (1)) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient						Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
U17	QA1 Parallel setup time, in phase, T_S (1)	2,12,21, 30,39	26,27,28,29	Open	22	21	72	416	584	900	1,100	ns	18,19
U18	QB1				23								
U19	QC1				24								
U20	QD1		26,27,28,29		25								
U21	QA2		3,4,5,6		7								
U22	QB2				8								
U23	QC2				9								
U24	QD2		3,4,5,6		11								
U25	QA3		35,36,37,38		31								
U26	QB3				32								
U27	QC3				33								
U28	QD3		35,36,37,38		34								
U29	QA4		13,14,15,16		17								
U30	QB4				18								
U31	QC4				19								
U32	QD4 Parallel setup time, in phase, T_S (1)	2,12,21, 30,39	13,14,15,16	Open	20	21	72	416	584	900	1,100	ns	18,19

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TABLE XVII. Parallel hold time (out of phase, T_H (0) and T_H (1)) test.

Test condition		Parameter	Connect				Initial limits 25° ±5°C Ambient						Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
							Min	Max	Min	Max	Min	Max		
V1	QA1	Parallel hold time, out of phase, T_H (0)	2,12,21, 30,39	26,27,28,29	Open	22	450	603	416	584	900	1,100	ns	18,20
V2	QB1					23								
V3	QC1					24								
V4	QD1			26,27,28,29		25								
V5	QA2			3,4,5,6		7								
V6	QB2					8								
V7	QC2					9								
V8	QD2			3,4,5,6		11								
V9	QA3			35,36,37,38		31								
V10	QB3					32								
V11	QC3					33								
V12	QD3			35,36,37,38		34								
V13	QA4			13,14,15,16		17								
V14	QB4					18								
V15	QC4					19								
V16	QD4	Parallel hold time, out of phase, T_H (0)	2,12,21, 30,39	13,14,15,16	Open	20	450	603	416	584	900	1,100	ns	18,20

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* TABLE XVII. Parallel hold time (out of phase, T_H (0) and T_H (1)) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° +5°C Ambient						Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
V17	QA1 Parallel hold time, out of phase, T_H (1)	2,12,21, 30,39	26,27,28,29	Open	22	450	603	416	584	900	1,100	ns	18,20
V18	QB1	↓	↓	↓	23	↓	↓	↓	↓	↓	↓	↓	↓
V19	QC1	↓		↓	24	↓	↓	↓	↓	↓	↓	↓	↓
V20	QD1	↓	26,27,28,29	↓	25	↓	↓	↓	↓	↓	↓	↓	↓
V21	QA2	↓	3,4,5,6	↓	7	↓	↓	↓	↓	↓	↓	↓	↓
V22	QB2	↓	↓	↓	8	↓	↓	↓	↓	↓	↓	↓	↓
V23	QC2	↓		↓	9	↓	↓	↓	↓	↓	↓	↓	↓
V24	QD2	↓	3,4,5,6	↓	11	↓	↓	↓	↓	↓	↓	↓	↓
V25	QA3	↓	35,36,37,38	↓	31	↓	↓	↓	↓	↓	↓	↓	↓
V26	QB3	↓	↓	↓	32	↓	↓	↓	↓	↓	↓	↓	↓
V27	QC3	↓		↓	33	↓	↓	↓	↓	↓	↓	↓	↓
V28	QD3	↓	35,36,37,38	↓	34	↓	↓	↓	↓	↓	↓	↓	↓
V29	QA4	↓	13,14,15,16	↓	17	↓	↓	↓	↓	↓	↓	↓	↓
V30	QB4	↓	↓	↓	18	↓	↓	↓	↓	↓	↓	↓	↓
V31	QC4	↓		↓	19	↓	↓	↓	↓	↓	↓	↓	↓
V32	QD4 Parallel hold time, out of phase, T_H (1)	2,12,21, 30,39	13,14,15,16	Open	20	450	603	416	584	900	1,100	ns	18,20

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TABLE XVIII. Serial setup time (out of phase, T_S (0) and T_S (1)) test.

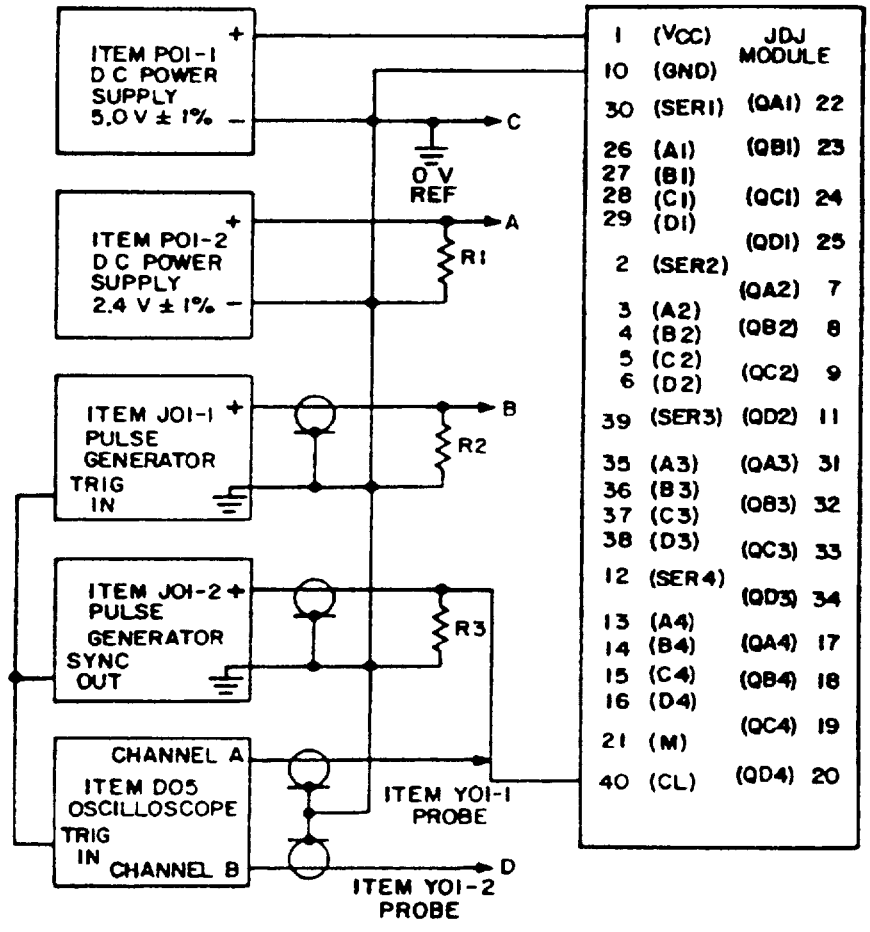
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient						Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
W1	Serial setup time, out of phase, T_S (0)	Open	30	21,26,27,28,29	25	471	622	416	584	900	1,100	ns	18,21
W2	↓	↓	2	3,4,5,6,21	11	↓	↓	↓	↓	↓	↓	↓	↓
W3	↓	↓	39	21,35,36,37,38	34	↓	↓	↓	↓	↓	↓	↓	↓
W4	Serial setup time, out of phase, T_S (0)	↓	12	13,14,15,16,21	20	↓	↓	↓	↓	↓	↓	↓	↓
W5	Serial setup time, out of phase, T_S (1)	↓	30	21,26,27,28,29	25	↓	↓	↓	↓	↓	↓	↓	↓
W6	↓	↓	2	3,4,5,6,21	11	↓	↓	↓	↓	↓	↓	↓	↓
W7	↓	↓	39	21,35,36,37,38	34	↓	↓	↓	↓	↓	↓	↓	↓
W8	Serial setup time, out of phase, T_S (1)	Open	12	13,14,15,16,21	20	471	622	416	584	900	1,100	ns	18,21

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TABLE XIX. Serial hold time (in phase, T_H (0) and T_H (1)) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient						Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	TA		TB		TC			
						Min	Max	Min	Max	Min	Max		
X1	Serial hold time, inphase, T_H (0)	Open	30	21,26,27,28,29	25	0	53	416	584	900	1,100	ns	18,22
X2	↓		2	3,4,5,6,21	11								
X3	↓		39	21,35,36,37,38	34								
X4	Serial hold time, in phase, T_H (0)		12	13,14,15,16,21	20								
X5	Serial hold time, in phase, T_H (1)		30	21,26,27,28,29	25								
X6	↓		2	3,4,5,6,21	11								
X7	↓		39	21,35,36,37,38	34								
X8	Serial hold time, in phase, T_H (1)	Open	12	13,14,15,16,21	20	0	53	416	584	900	1,100	ns	18,22

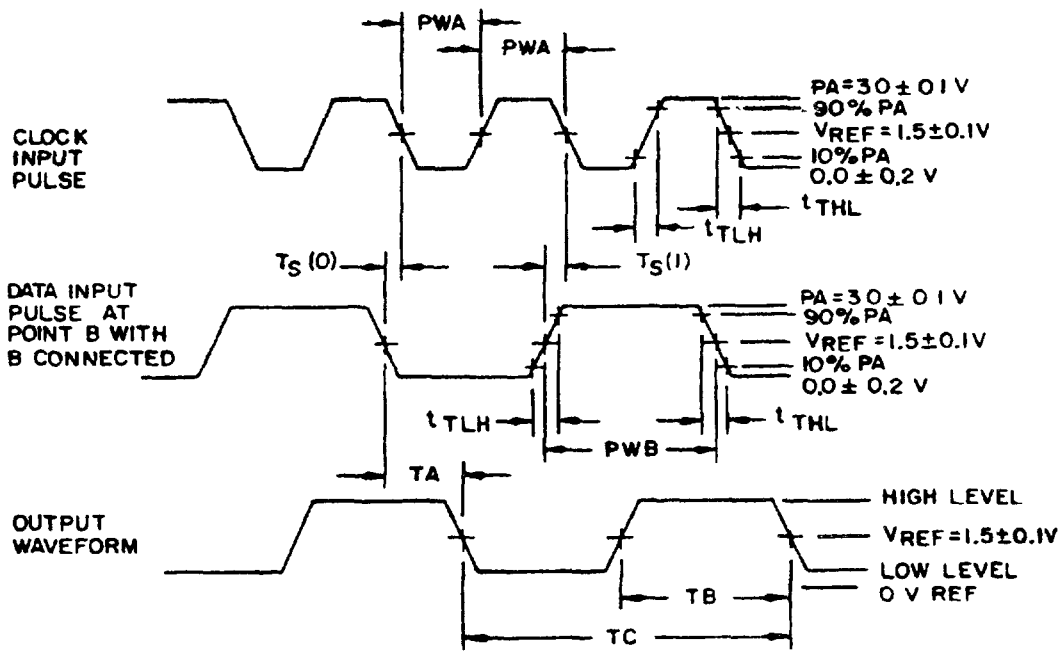
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NOTE. R1-R3 = 51Ω ±5%, 1/2 W.

* FIGURE 18. Parallel setup and hold time and serial setup and hold time test circuit for tables XVI, XVII, XVIII, and XIX.

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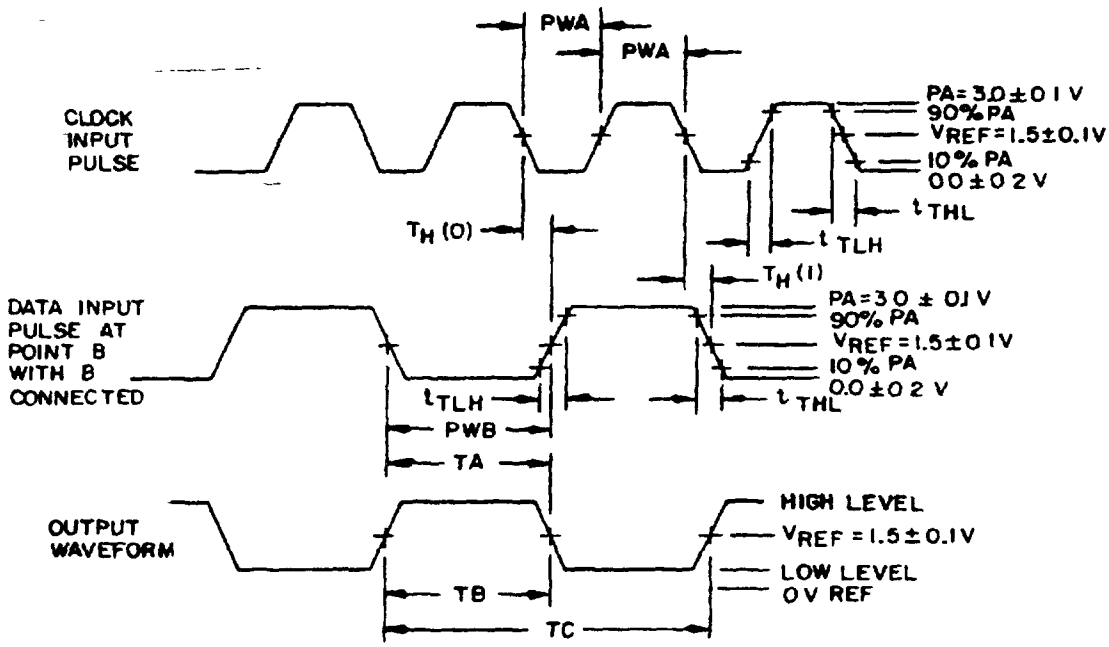
NOTES:
1. Input pulse parameters

   a. PWA = 250 ns ±10%.
   b. PWB = 500 ns ±10%.
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   d. For test conditions U1-U16, $T_S(0) = 20 \pm 1$ ns and $T_S(1) = 40 \pm 4$ ns.
   e. For test conditions U17-U32, $T_S(0) = 40 \pm 4$ ns and $T_S(1) = 20 \pm 1$ ns.

2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

* FIGURE 19. Parallel setup time (inphase, $T_S(0)$ and $T_S(1)$) test circuit waveforms for table XVI.

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NOTES

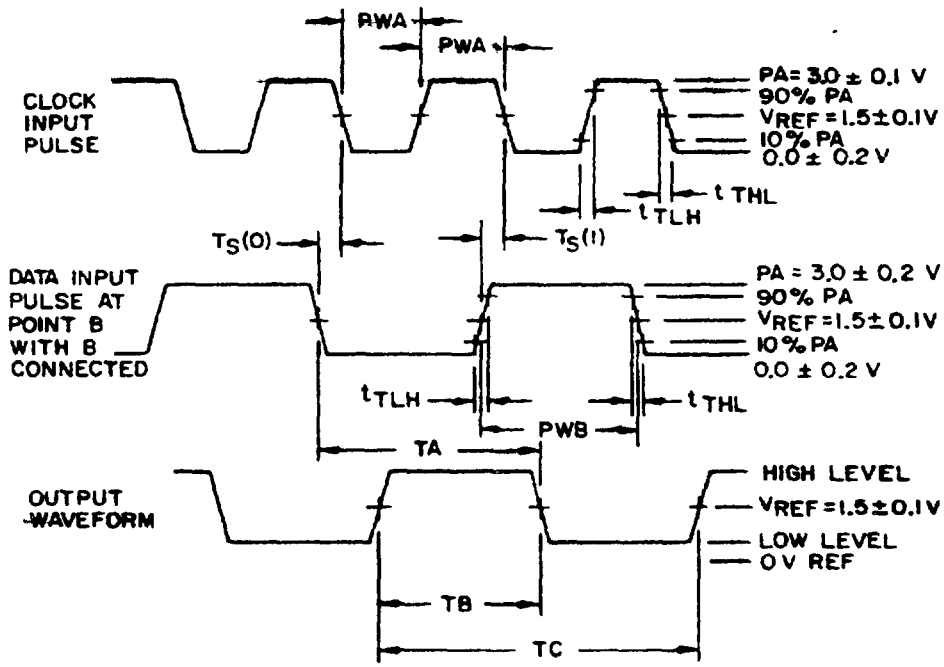
1. Input pulse parameters

   a. PWA = 250 ns ±10%.
   b. PWB = 500 ns ±10%.
   c. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   d. For test conditions, V1-V16, $T_H(0)$ = 0 ±2 ns and $T_H(1)$ = 10 ±1 ns.
   e. For test conditions, V17-V32, $T_H(0)$ = 10 ±1 ns and $T_H(1)$ = 0 ±2 ns.

2. Before making the measurements, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

* FIGURE 20. Parallel hold time (out of phase, $T_H(0)$ and $T_H(1)$) test circuit waveforms for table XVII.

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NOTES:
1. Input pulse parameters.

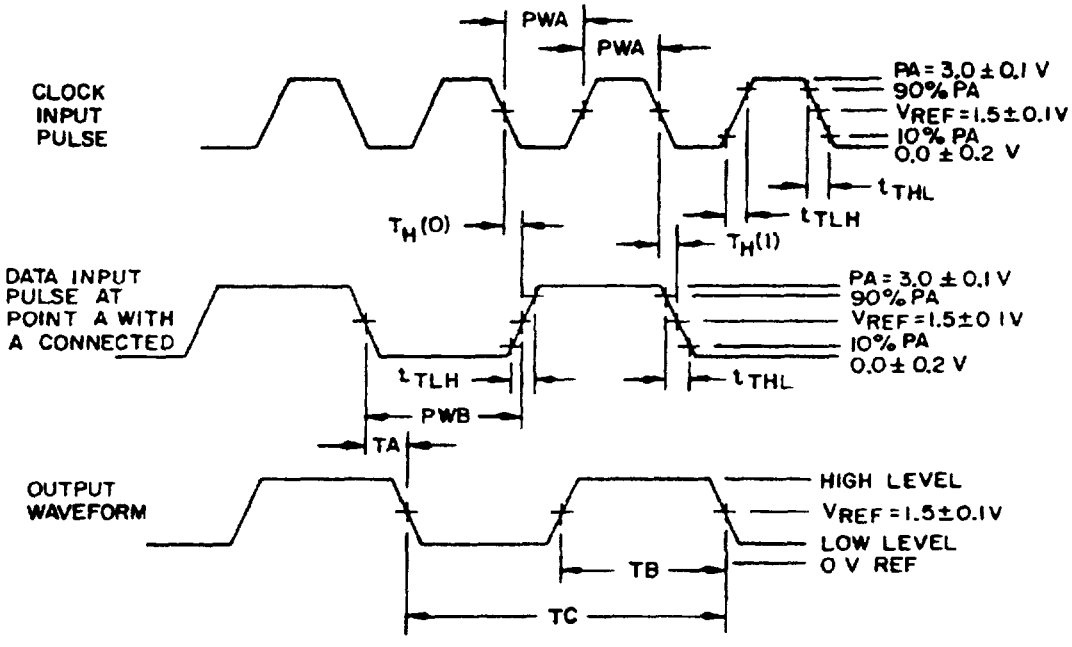
   a. PWA = 250 ns ±10%.
   b. PWB = 500 ns ±10%.
   c. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   d. For test conditions W1-W4, $T_S(0) = 20 \pm 1$ ns and $T_S(1) = 40 \pm 4$ ns.
   e. For test conditions W5-W8, $T_S(0) = 40 \pm 4$ ns and $T_S(1) = 20 \pm 1$ ns.

2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

*

FIGURE 21. Serial setup time (out of phase, $T_S(0)$ and $T_S(1)$) test circuit waveforms for table XVIII.

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NOTES
1. Input pulse parameters

   a. PWA = 250 ns ±10%.
   b. PWB = 500 ns ±10%.
   c $t_{TLH} = t_{THL}$ = 10 ±2 ns.
   d. For test conditions X1-X4, $T_H(0)$ = 0 ±2 ns and $T_H(1)$ = 10 ±1 ns.
   e. For test conditions X5-X8, $T_H(0)$ = 10 ±1 ns and $T_H(1)$ = 0 ±2 ns.

2. Before making the measurements, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.4 V.

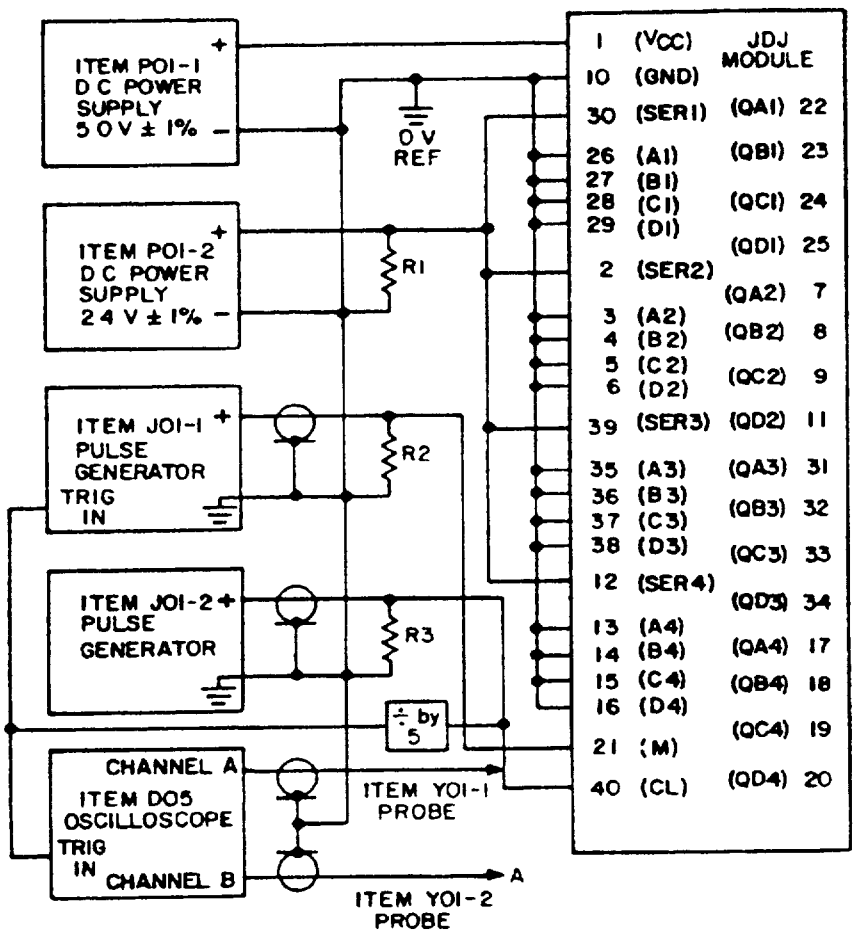
* FIGURE 22. Serial hold time (in phase, $T_H(0)$ and $T_H(1)$) test circuit waveforms for table XIX.

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TABLE XX. Mode control setup time test.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient						Unit	Test figure
			TA		TB		TC			
			Min	Max	Min	Max	Min	Max		
Y1	Mode control setup time	22	210	283	118	202	380	420	ns	23,24
Y2		7								
Y3		31								
Y4	Mode control setup time	17	210	283	118	202	380	420	ns	23,24

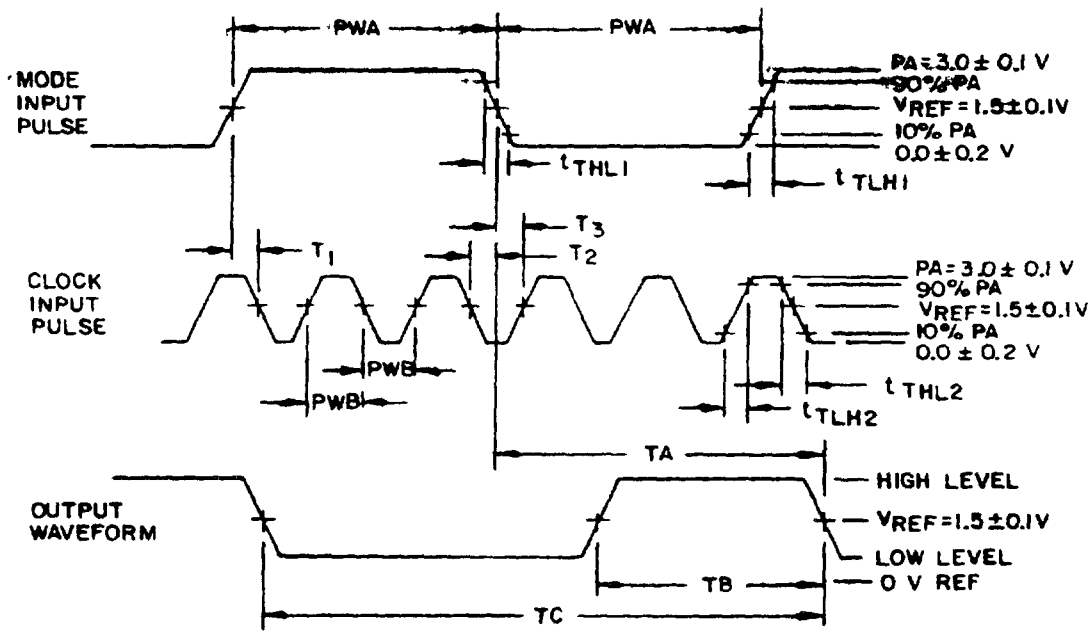
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NOTE: R1-R3 = 51Ω ±5%, 1/2 W.

FIGURE 23. Mode control setup time test circuit for table XX.

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NOTES
1. Input pulse parameters.

   a. PWA = 200 ns ±5%
   b. PWB = 40 ns ±5%.
   c. $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   d. $t_{TLH2} = t_{THL2} = 10 +0, -2$ ns.
   e. $T_1 = T_2 = T_3 = 20 \pm 2$ ns.

2. Before making the measurements, verify that the output pulse high level is > 2.4 V and the low level is ≤ 0.4 V.

* FIGURE 24. Mode control setup time test circuit waveforms for table XX.

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TABLE XXI. Output crosstalk (low-level and duration) test

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	QA1	Output crosstalk, low level	Open	3,4,5,6,13,14,15, 16,27,28,29,35,36, 37,38	26	22	0	800	mV	25,26
Z2	QB1			3,4,5,6,13,14,15, 16,26,28,29,35,36, 37,38	27	23				
Z3	QC1			3,4,5,6,13,14,15, 16,26,27,29,35,36, 37,38	28	24				
Z4	QD1			3,4,5,6,13,14,15, 16,26,27,28,35,36, 37,38	29	25				
Z5	QA2			4,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38	3	7				
Z6	QB2			3,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38	4	8				
Z7	QC2			3,4,6,13,14,15,16, 26,27,28,29,35,36, 37,38	5	9				
Z8	QD2			3,4,5,13,14,15,16, 26,27,28,29,35,36, 37,38	6	11				
Z9	QA3	Output crosstalk, low level	Open	3,4,5,6,13,14,15, 16,26,27,28,29,36, 37,38	35	31	0	800	mV	25,26

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TABLE XXI. Output crosstalk (low-level and duration) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Z10	QB3	Output crosstalk, low level	Open	3,4,5,6,13,14,15, 16,26,27,28,29,35, 37,38	36	32	0	800	mV	25,26
Z11	QC3			3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,38	37	33				
Z12	QD3			3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,37	38	34				
Z13	QA4			3,4,5,6,14,15,16, 26,27,28,29,35,36, 37,38	13	17				
Z14	QB4			3,4,5,6,13,15,16, 26,27,28,29,35,36, 37,38	14	18				
Z15	QC4			3,4,5,6,13,14,16, 26,27,28,29,35,36, 37,38	15	19				
Z16	QD4	Output crosstalk, low level		3,4,5,6,13,14,15, 26,27,28,29,35,36, 37,38	16	20		800		25,26
Z17	QA1	Output crosstalk, duration		3,4,5,6,13,14,15, 16,27,28,29,35,36, 37,38	26	22		400		25,27
Z18	QB1	Output crosstalk, duration	Open	3,4,5,6,13,14,15, 16,26,28,29,35,36, 37,38	27	23	0	400	mV	25,27

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TABLE XXI. Output crosstalk (low-level and duration) test - Continued.

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Z19	QC1	Output crosstalk, duration	Open	3,4,5,6,13,14,15, 16,26,27,29,35,36, 37,38	28	24	0	400	mV	25,27
Z20	QD1			3,4,5,6,13,14,15, 16,26,27,28,35,36, 37,38	29	25				
Z21	QA2			4,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38	3	7				
Z22	QB2			3,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38	4	8				
Z23	QC2			3,4,6,13,14,15,16, 26,27,28,29,35,36, 37,38	5	9				
Z24	QD2			3,4,5,13,14,15,16, 26,27,28,29,35,36, 37,38	6	11				
Z25	QA3			3,4,5,6,13,14,15, 16,26,27,28,29,36, 37,38	35	31				
Z26	QB3			3,4,5,6,13,14,15, 16,26,27,28,29,35, 37,38	36	32				
Z27	QC3	Output crosstalk, duration	Open	3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,38	37	33	0	400	mV	25,27

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TABLE XXI. Output crosstalk (low-level and duration) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z28	QD3 Output crosstalk, duration	Open	3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,37	38	34	0	400	mV	25,27
Z29	QA4		3,4,5,6,14,15,16, 26,27,28,29,35,36, 37,38	13	17				
Z30	QB4		3,4,5,6,13,15,16, 26,27,28,29,35,36, 37,38	14	18				
Z31	QC4		3,4,5,6,13,14,16, 26,27,28,29,35,36, 37,38	15	19				
Z32	QD4 Output crosstalk, duration	Open	3,4,5,6,13,14,15, 26,27,28,29,35,36, 37,38	16	20	0	400	mV	25,27

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TABLE XXII. Output crosstalk (high-level) test.

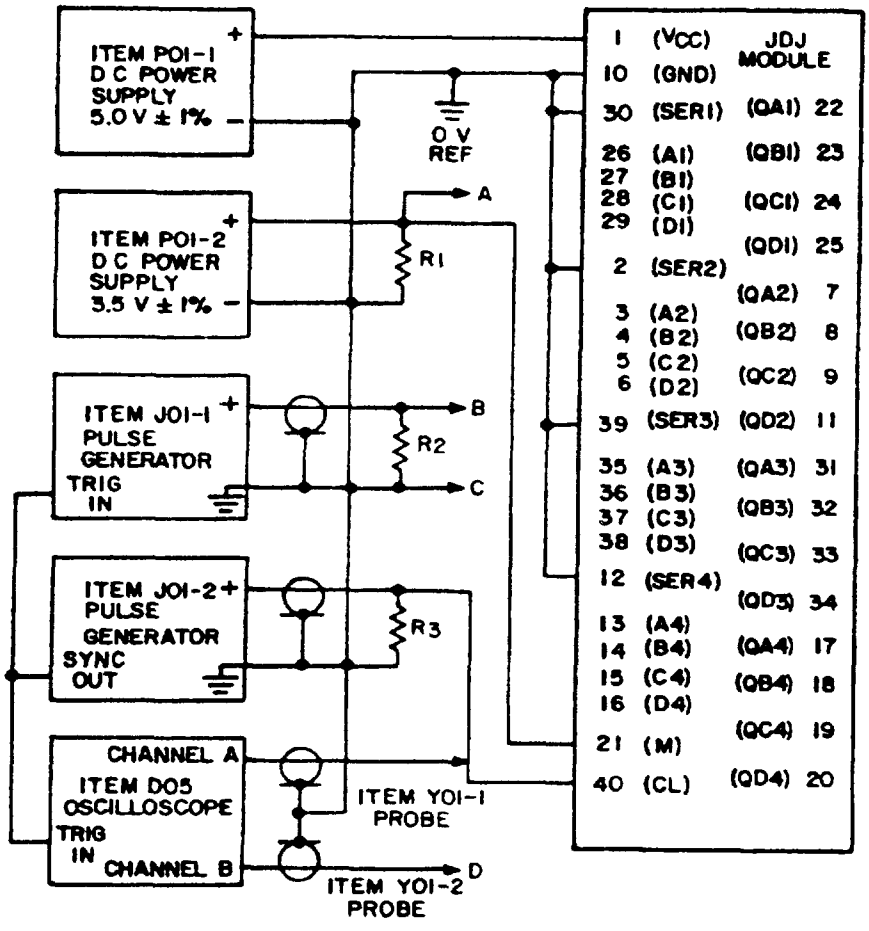
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	QA1 Output crosstalk, high level	26	3,4,5,6,13,14,15, 16,27,28,29,35,36, 37,38	Open	22	0	500	mV	25,26
AA2	QB1	27	3,4,5,6,13,14,15, 16,26,28,29,35,36, 37,38		23				
AA3	QC1	28	3,4,5,6,13,14,15, 16,26,27,29,35,36, 37,38		24				
AA4	QD1	29	3,4,5,6,13,14,15, 16,26,27,28,35,36, 37,38		25				
AA5	QA2	3	4,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38		7				
AA6	QB2	4	3,5,6,13,14,15,16, 26,27,28,29,35,36, 37,38		8				
AA7	QC2	5	3,4,6,13,14,15,16, 26,27,28,29,35,36, 37,38		9				
AA8	QD2	6	3,4,5,13,14,15,16, 26,27,28,29,35,36, 37,38		11				
AA9	QA3 Output crosstalk, high level	35	3,4,5,6,13,14,15, 16,26,27,28,29,36, 37,38	Open	31	0	500	mV	25,26

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TABLE XXII. Output crosstalk (high-level) test Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA10	QB3 Output crosstalk, high level	36	3,4,5,6,13,14,15, 16,26,27,28,29,35, 37,38	Open	32	0	500	mV	25,26
AA11	QC3	37	3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,38		33				
AA12	QD3	38	3,4,5,6,13,14,15, 16,26,27,28,29,35, 36,37		34				
AA13	QA4	13	3,4,5,6,14,15,16, 26,27,28,29,35,36, 37,38		17				
AA14	QB4	14	3,4,5,6,13,15,16, 26,27,28,29,35,36, 37,38		18				
AA15	QC4	15	3,4,5,6,13,14,16, 26,27,28,29,35,36, 37,38		19				
AA16	QD4 Output crosstalk,	16	3,4,5,6,13,14,15, 26,27,28,29,35,36, 37,38	Open	20	0	500	mV	25,26

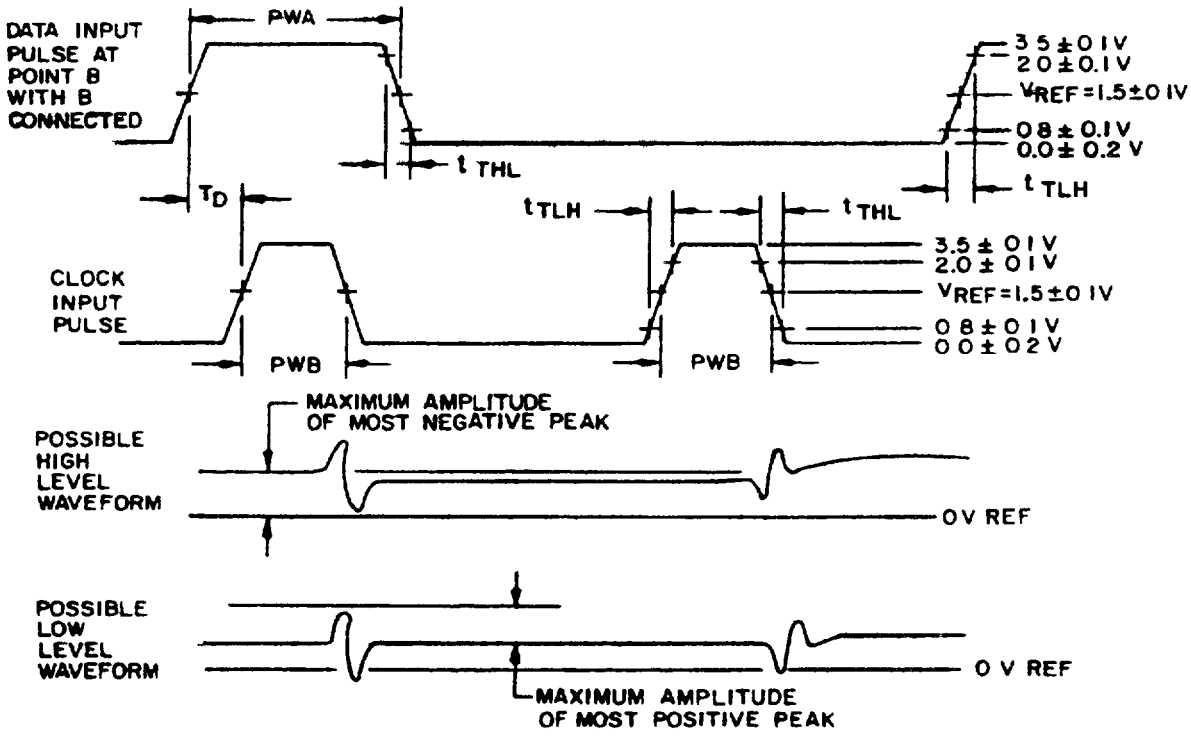
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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as specified on figure 10.

FIGURE 25. Output crosstalk (low-level, high-level, and duration) test circuit for tables XXI and XXII.

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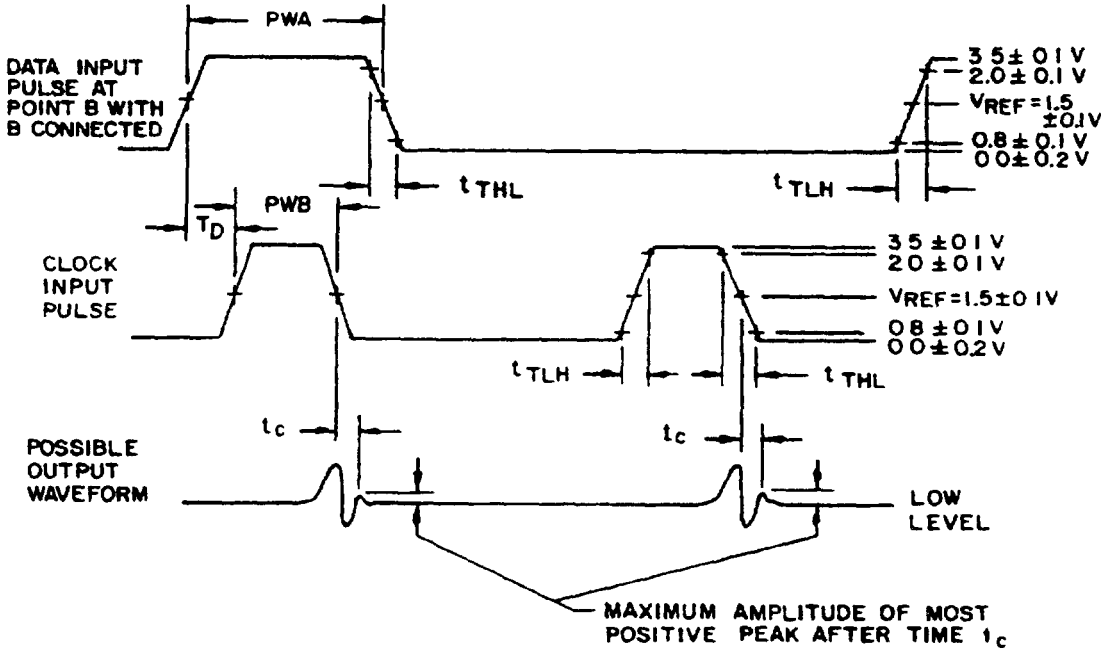


NOTE Input pulse parameters:

a. PWA = 200 ns ±10%.
b. PWB = 100 ns ±10%.
c. Data PRR = 1.0 MHz ±10%.
d. Clock PRR = 2.0 MHz ±10%.
e. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
f. $T_D = 50$ ns ±10%.

* FIGURE 26. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXI and XXII.

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NOTE. Input pulse parameters:

a. PWA = 200 ns ±10%.
b. PWB = 100 ns ±10%.
c. Data PRR = 1.0 MHz ±10%.
d. Clock PRR = 2.0 MHz ±10%.
e. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
f. T_D = 50 ns ±10%.
g. t_c = 51 ns.

*

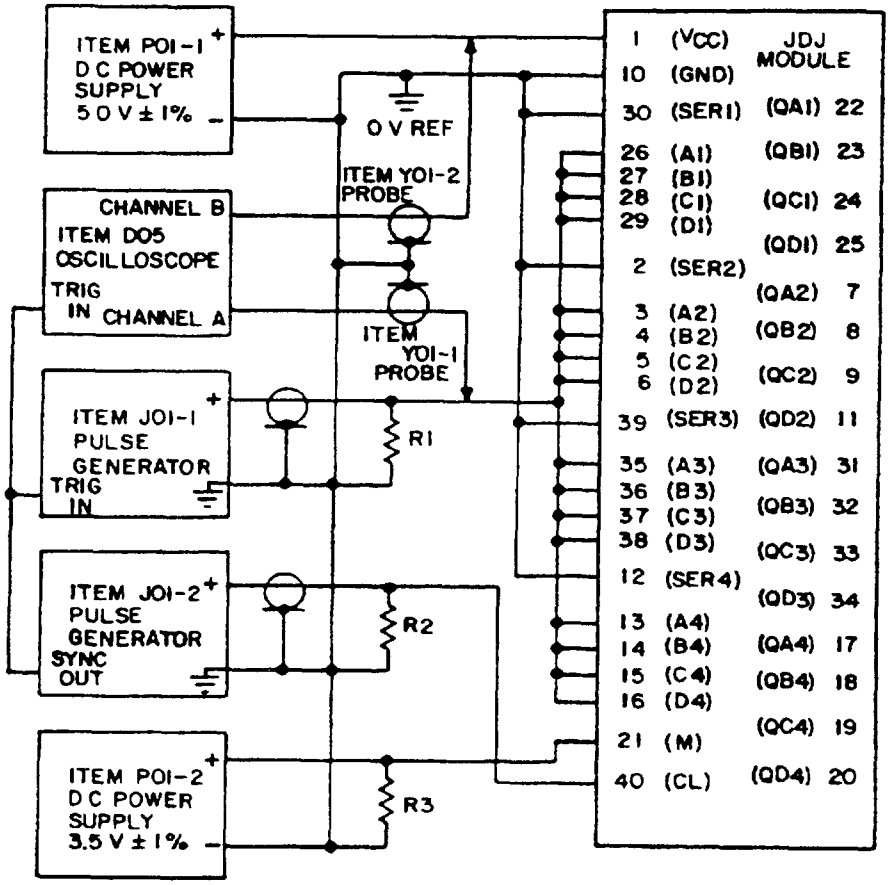
FIGURE 27. Output crosstalk (duration) test circuit waveforms for table XXI

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TABLE XXIII. Power supply transient amplitude test

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
BB	Power supply transient amplitude	0	400	mV	28,29

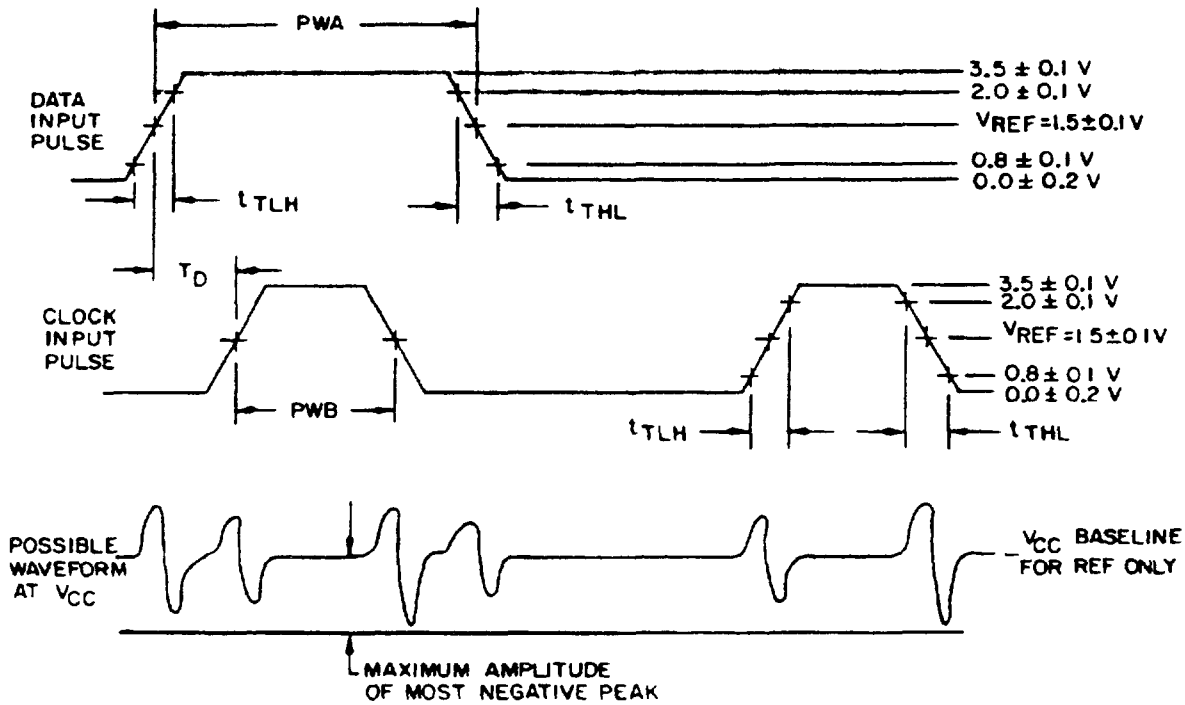
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:36Z
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NOTE: R1-R3 = 51Ω ±5%, 1/2 W.

FIGURE 28. Power supply transient amplitude test circuit for table XXIII.

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NOTE: Input pulse parameters

a. PWA = 3 µs ±10%.
b. PWB = 1.5 µs ±10%.
c. Data PRR = 100 kHz ±10%.
d. Clock PRR = 200 kHz ±10%.
e. t_{TLH} = t_{THL} = 3.5 ±1.5 ns (verified at each input pin).
f. T_D = 750 ns ±10%.

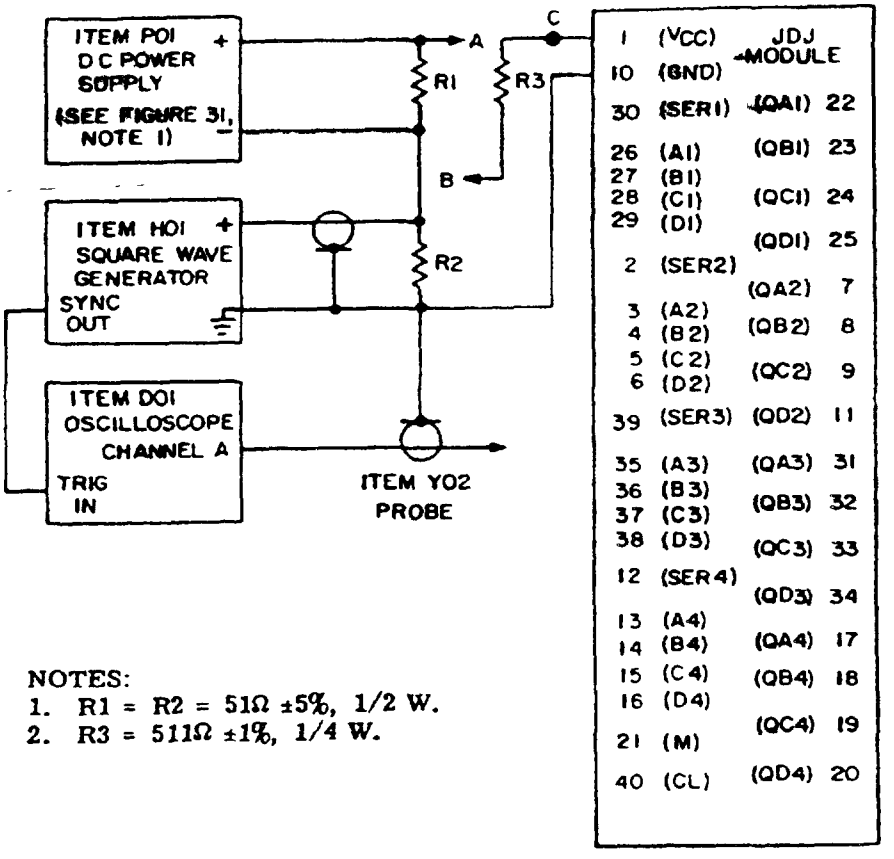
* FIGURE 29. Power supply transient amplitude test circuit waveforms for table XXIII.

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TABLE XXIV. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
CC	Power supply transient rise time	1.0	2.0	ms	30,31

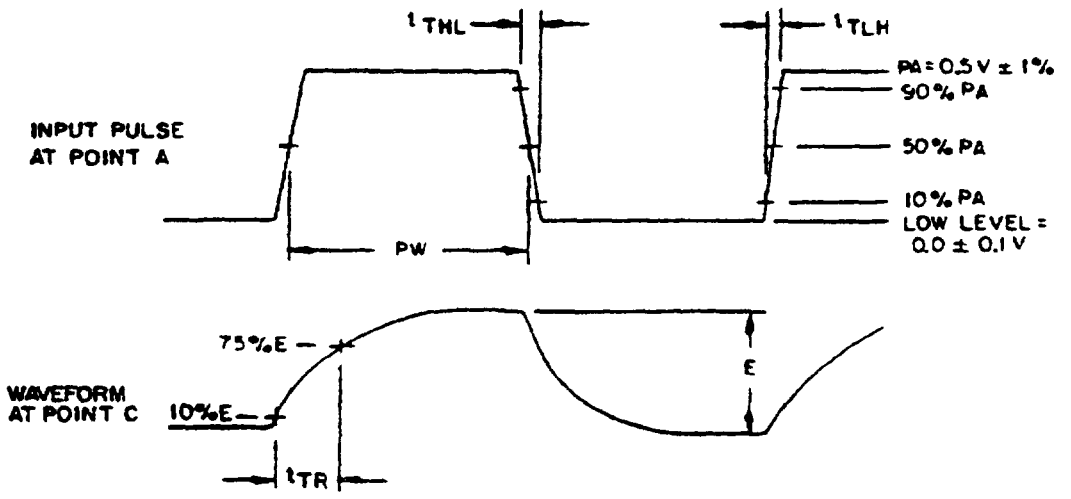
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 30. Power supply transient rise time test circuit for table XXIV.

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NOTES:
1. Input pulse parameters with point A open:
   a. PW = 10 ms ±10%.
   b. PRR = 50 Hz (reference).
   c. $t_{TLH} = t_{THL} \leq 100$ ns (reference).
   d. Adjust item P01 for the specified low level.
2. Connect point A to point B and recheck the input pulse parameters at point A.
3. Measure transient rise time (t_{TR}) at point C.

FIGURE 31. Power supply transient rise time test circuit waveforms for table XXIV.

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TABLE XXV. Input clamp diode voltage test

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
DD1	SER1 Input clamp diode voltage	30	-0.4	-1.5	V	32
DD2	A1	26				
DD3	B1	27				
DD4	C1	28				
DD5	D1	29				
DD6	SER2	2				
DD7	A2	3				
DD8	B2	4				
DD9	C2	5				
DD10	D2	6				
DD11	SER3	39				
DD12	A3	35				
DD13	B3	36				
DD14	C3	37				
DD15	D3	38				
DD16	SER4	12				
DD17	A4	13				
DD18	B4 Input clamp diode voltage	14	-0.4	-1.5	V	32

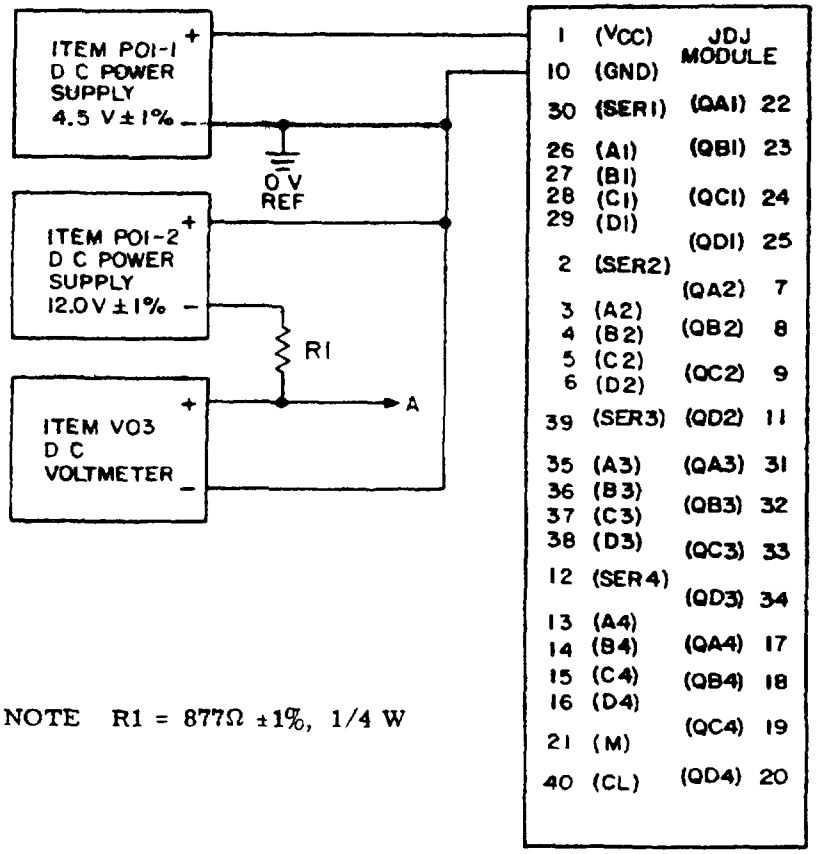
MIL-M-38510/328A(USAF)

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TABLE XXV. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
DD 9	C4 Input clamp diode voltage	15	-0.4	-1.5	V	32
DD10	D4	16				
DD11	M	21				
DD12	C1. Input clamp diode voltage	40	-0.4	-1 5	V	32

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NOTE R1 = 877Ω ±1%, 1/4 W

FIGURE 32 Input clamp diode voltage test circuit for table XXV.

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TABLE XXVI. False shift immunity test.

Test condition	Parameter	Connect			Output logic states				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	QA	QB	QC	QD	Min	Max		
EE1	False shift immunity	30	Open	26,27,28,29	L	L	L	L	See figure 33, notes 4 and 8			33
EE2		2		3,4,5,6	L	L	L	L				
EE3		39		35,36,37,38	L	L	L	L				
EE4		12		13,14,15,16	L	L	L	L	See figure 33, notes 4 and 8			
EE5		27,28,29,30		26	H	L	L	L	See figure 33, notes 5 and 8			
EE6		26,28,29,30		27	L	H	L	L				
EE7		26,27,29,30		28	L	L	H	L				
EE8		26,27,28,30		29	L	L	L	H				
EE9		2,4,5,6		3	H	L	L	L				
EE10		2,3,5,6		4	L	H	L	L				
EE11		2,3,4,6		5	L	L	H	L				
EE12		2,3,4,5		6	L	L	L	H				
EE13		36,37,38,39		35	H	L	L	L				
EE14		35,37,38,39		36	L	H	L	L				
EE15		35,36,38,39		37	L	L	H	L				
EE16		35,36,37,39		38	L	L	L	H				
EE17		12,14,15,16		13	H	L	L	L				
EE18		12,13,15,16		14	L	H	L	L				
EE19	False shift immunity	12,13,14,16	Open	15	L	L	H	L	See figure 33, notes 5 and 8			33

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TABLE XXVI. False shift immunity test - Continued

Test condition	Parameter	Connect			Output logic states				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	QA	QB	QC	QD	Min	Max		
EE20	False shift immunity	12,13,14,15	Open	16	L	L	L	H	See figure 33, notes 5 and 8			33
EE21		26,28	27,29,30	Open	L	H	L	H	See figure 33, notes 6 and 8			
EE22		3,5	2,4,6		L	H	L	H				
EE23		35,37	36,38,39		L	H	L	H				
EE24		13,15	12,14,16		L	H	L	H	See figure 33, notes 6 and 8			
EE25		27,29,30	26,28		H	L	H	L	See figure 33, notes 7 and 8			
EE26		2,4,6	3,5		H	L	H	L				
EE27		36,38,39	35,37		H	L	H	L				
EE28	False shift immunity	12,14,16	13,15	Open	H	L	H	I	See figure 33, notes 7 and 8			33

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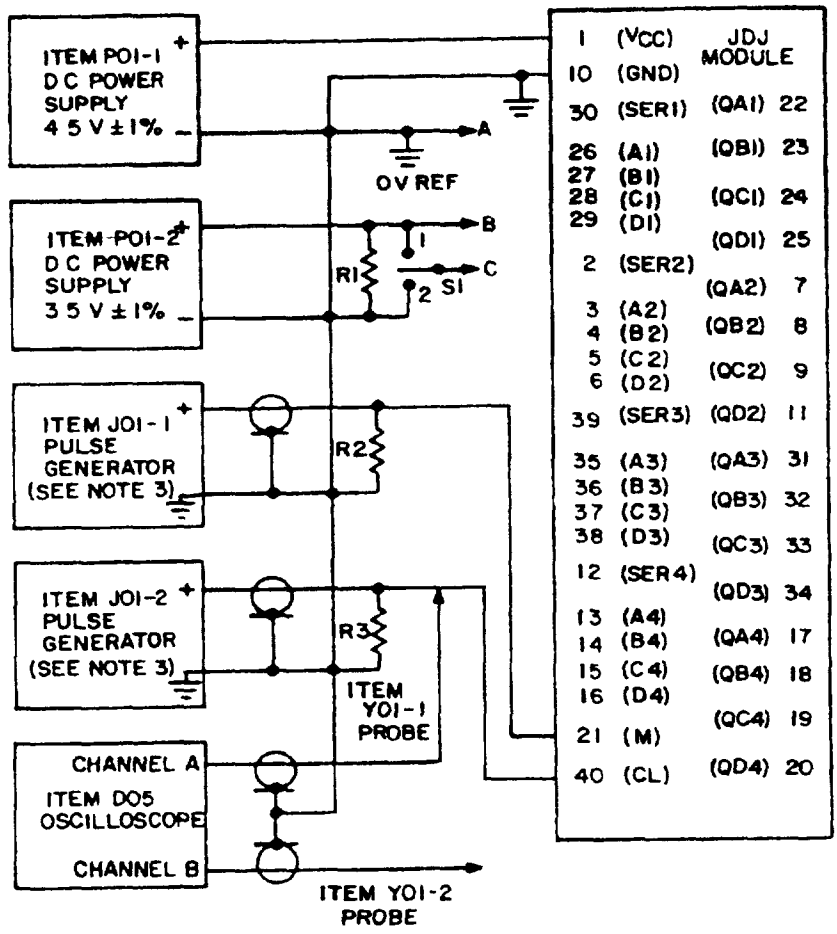
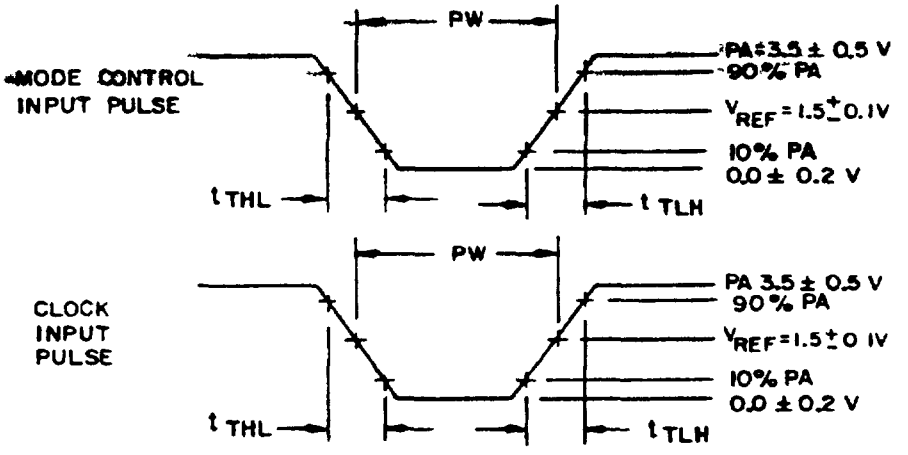


FIGURE 33. False shift immunity test circuit for table XXVI.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. S1 = SPDT.
3. Input pulse parameters·



a. PW = 500 ns ±10%.
b. PRR = 500 kHz ±50%.
c. $t_{TLH} = t_{THL}$ = 100 ns ±10%.
d. After adjusting pulse generators J01-1 and J01-2 for the above waveforms, place both J01-1 and J01-2 in the single-pulse mode.

4. Test procedure for test conditions EE1-EE4:

a. With switch S1 in position 2, make all connections for test condition EE1.
b. Apply one clock pulse (J01-2) to the circuit under test.
c. Place switch S1 in position 1.
d. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI.
e. Repeat the above procedure for test conditions EE2-EE4.

* FIGURE 33. False shift immunity test circuit for table XXVI - Continued.

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5. Test procedure for test conditions EE5-EE20:

a. With switch S1 in position 1, make all connections for test condition EE5.
b. Apply one clock pulse (J01-2) to the circuit under test.
c. Place switch S1 in position 2.
d. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI
e. With switch S1 in position 1, make all connections for test condition EE6.
f. Apply one clock pulse (J01-2) to the circuit under test.
g. Place switch S1 in position 2.
h. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI.
i. With switch S1 in position 1, make all connections for test condition EE7.
j. Apply one clock pulse (J01-2) to the circuit under test.
k. Place switch S1 in position 2.
l. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI
m. With switch S1 in position 1, make all connections for test condition EE8.
n. Apply one clock pulse (J01-2) to the circuit under test.
o. Place switch S1 in position 2.
p. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI
q. Repeat the above procedure for test conditions EE9-EE12, EE13-EE16, and EE17-EE20.

6. Test procedure for test conditions EE21-EE24

a. Make all connections for test condition EE21.
b. Apply one clock pulse (J01-2) to the circuit under test.
c. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI.
d. Repeat the above procedure for test conditions EE22-EE24.

7. Test procedure for test conditions EE25-EE28

a. Make all connections for test condition EE25.
b. Apply one clock pulse (J01-2) to the circuit under test.
c. Apply one mode pulse (J01-1) to the circuit under test and verify that the output logic states are as specified in table XXVI.
d. Repeat the above procedure for test conditions EE26-EE28.

8. Output logic states:

a. H = High level = 2.4 V minimum, 4.5 V maximum.
b. L = Low level = 0.0 V minimum, 0.4 V maximum.

* FIGURE 33. False shift immunity test circuit for table XXVI - Continued.

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